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PAPER

Spectral properties of the Riemann zeta function and their physical applications

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Abstract

This work studies the Riemann ζ numerically with emphasis on its physical manifestations. Three areas are examined: (i) the spectral statistics of the non-trivial zeros, which we compare with the Gaussian Unitary Ensemble predictions, obtaining $\chi^2/\mathrm{dof} = 1.2$; (ii) the role of $\zeta(\frac{3}{2})$ in determining the critical temperature for Bose-Einstein condensation, where we find $T_c = 0.13$ K for an ideal gas; and (iii) the application of ζ regularisation to the Casimir effect, yielding the standard $-\frac{\pi^2\hbar c}{720a^3}$ energy, which for $a = 1$ nm gives $\sim -4.33 \times 10^{-4}$ J/m^2. The numerical results are consistent with the Montgomery-Odlyzko law and with spectral interpretations associated with the Hilbert-Pólya conjecture, within the statistical limitations of the analysis.

1 Introduction

The Riemann zeta function, $\zeta(s)$[1], is central to analytic number theory and arises in several branches of physics, namely quantum chaos [1], statistical mechanics [2], quantum field theory (QFT) [3], and cosmological models [4]. We study numerically three physical manifestations of ζ - the spectral statistics of its non-trivial zeros, the critical temperature for Bose-Einstein condensation (BEC), and the Casimir effect, which depend on the values $\zeta(\frac{3}{2})$ and $\zeta(-3)$, respectively. For $\Re(s) > 1$, the Dirichlet series [5]

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - p^{-s}\right)^{-1}, \tag{1}$$

converges absolutely. The analytic continuation of $\zeta(s)$ to the whole complex plane, due to Riemann [5], is given by the integral representation

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1}\,\mathrm{d}x = \frac{1}{\Gamma(s)} \sum_{n=1}^{\infty} \int_0^\infty x^{s-1} e^{-nx}\,\mathrm{d}x, \tag{2}$$

and the functional equation [6]

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\,\zeta(1-s), \qquad s \in \mathbb{C}, \tag{3}$$

which imposes a symmetry about the critical line $\Re(s) = \frac{1}{2}$. The non-trivial zeros $\rho = \frac{1}{2} + i\gamma$ with $\gamma \in \mathbb{R}$, are the subject of the Riemann hypothesis (RH) as all such zeros lie on this line [5]. Here $\Gamma(z)$ denotes the Euler gamma function $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\,\mathrm{d}t$ for $\Re(z) > 0$, extended analytically elsewhere. The functional equation also implies that the zeros are symmetric about the critical line and appear in complex conjugate pairs. For numerical work on the critical line, the Riemann-Siegel (RS) formula [7] gives the approximation

$$\zeta\left(\frac{1}{2} + it\right) = \sum_{n \le x} \frac{1}{n^{\frac{1}{2}+it}} + \chi\left(\frac{1}{2} + it\right) \sum_{n \le y} \frac{1}{n^{\frac{1}{2}-it}} + \mathcal{O}\left(x^{-\frac{1}{2}} + |t|^{\frac{1}{2}} y^{-\frac{1}{2}}\right), \qquad xy = \frac{t}{2\pi}, \tag{4}$$

[1] From now on, we shall often write ζ, Γ, Z and θ for $\zeta(s)$, $\Gamma(s)$, $Z(t)$ and $\theta(t)$, respectively, when the argument is clear from the context. We shall also omit the condition $\Re(s) > 1$ where it is clear from the context.

with $xy = \frac{t}{2\pi}$ and $\chi(s) = \frac{\pi^{s-\frac{1}{2}}\Gamma(1-s)}{\Gamma(s)}$. The Hardy Z-function is defined as

$$Z(t) = e^{i\theta(t)}\zeta\left(\frac{1}{2}+it\right), Z(t) \in \mathbb{R}, \quad \forall t \in \mathbb{R}, \tag{5}$$

with $\theta(t)$ the RS theta function. This reduces the zero-finding problem to finding sign changes of the real function $Z(t)$. The Euler product [8]

$$\zeta(s) = \prod_p \left(1-p^{-s}\right)^{-1}, \tag{6}$$

relates ζ to the primes [8]. The logarithmic derivative gives

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s} = \sum_p\sum_{k=1}^{\infty}\frac{\log p}{p^{ks}}, \tag{7}$$

where $\Lambda(n) = \log p$ if $n = p^k$ and zero otherwise. This identity yields the explicit formula [6, 8]

$$\psi(x) = x - \sum_\rho \frac{x^\rho}{\rho} - \frac{1}{2}\log\left(1-x^{-2}\right) - \log(2\pi), \qquad x > 1, \tag{8}$$

with $\psi(x) = \sum_{n\le x}\Lambda(n)$. The RH is equivalent to the bound

$$\sum_{n\le x}\Lambda(n) = x + \mathcal{O}\left(x^{\frac{1}{2}+\varepsilon}\right) \tag{9}$$

for every $\varepsilon > 0$. Montgomery conjectured the GUE pair correlation [9]

$$R_2(u) = 1 - \left(\frac{\sin\pi u}{\pi u}\right)^2, \qquad u \in \mathbb{R}, \tag{10}$$

as verified numerically by Odlyzko [10]. This supports the Hilbert-Pólya conjecture (HPc) [11], that a self-adjoint operator has the zero imaginary parts as eigenvalues. The corresponding spacing distribution is the Wigner-Dyson form

$$P_{\mathrm{GUE}}(s) = \frac{\pi s}{2}\exp\left(-\frac{\pi s^2}{4}\right), \tag{11}$$

with level repulsion $P(s) \sim s$ as $s \to 0$, unlike the Poisson case $P(s) = e^{-s}$. In QFT, $\zeta(s)$ provides a rigorous framework for the regularisation of divergent sums [12, 13], a canonical example being the vacuum energy between two perfectly conducting plates separated by a distance a, which follows from the mode sum over all polarisations and transverse momenta [14, 15]

$$E_{\mathrm{v}}(a) = \sum_{n=1}^{\infty}\frac{\hbar c}{2}\sqrt{\left(\frac{\pi n}{a}\right)^2 + k_\perp^2}, \tag{12}$$

which zeta regularisation [3] yields as

$$E_C(a) = -\frac{\pi^2\hbar c}{720a^3}\left(\frac{\varepsilon_r-1}{\varepsilon_r+1}\right) = -\frac{\hbar c}{2a^3}\frac{\zeta(-3)}{\pi}\left(\frac{\varepsilon_r-1}{\varepsilon_r+1}\right), \qquad \zeta(-3) = \frac{1}{120}, \tag{13}$$

where ε_r is the relative permittivity. The factor $\frac{\pi^2}{720}$ follows from $\zeta(-3) = \frac{1}{120}$, which regularises the divergent sum $\sum n^3$. In statistical mechanics, the critical temperature for BEC [2] is determined by $\zeta(\frac{3}{2})$, the density of states in three dimensions, and the grand potential of an ideal Fermi gas, which are collected as

$$\begin{cases} T_c = \dfrac{2\pi\hbar^2}{mk_B}\sqrt[3]{\left(\dfrac{n}{\zeta(\frac{3}{2})}\right)^2}, \\ \displaystyle\int_0^\infty \frac{\sqrt{x}}{e^x-1}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}\zeta\left(\frac{3}{2}\right), \\ \Omega = -\dfrac{gV}{\lambda^3}k_BT\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}z^n}{n^{5/2}}, \end{cases} \tag{14}$$

where n is the density, $\lambda = \sqrt{2\pi\hbar^2/(mk_BT)}$ the thermal de Broglie wavelength, g the spin degeneracy, and $z = e^{\mu/(k_BT)}$ the fugacity. Zeros of ζ may be observed through classical wave scattering [16], with the reflection coefficient for a one-dimensional scattering system expressed in terms of ζ as

$$r(t) = \frac{\zeta(1/2+it)-1}{\zeta(1/2+it)+1}, \qquad Z(t) = \frac{1+r(t)}{1-r(t)} = \zeta(1/2+it), \tag{15}$$

where r has poles at the zeros γ_n of ζ, and Z diverges as $(t-\gamma_n)^{-1}$ near each zero. Near each zero γ_n, the reflection coefficient takes the Lorentzian form

$$r(t) \simeq \frac{i\eta_n}{(t-\gamma_n)+i\eta_n}, \quad |r(t)|^2 = \frac{\eta_n^2}{(t-\gamma_n)^2+\eta_n^2}, \quad \arg r(t) = \frac{\pi}{2} - \arctan\left(\frac{t-\gamma_n}{\eta_n}\right), \tag{16}$$

where $\eta_n = -\frac{1}{Z'(\gamma_n)}$ is the resonance width.

2 Methodology and computational framework

Our numerical analysis of ζ uses a custom Python 3.11 implementation, using `mpmath` for high-precision complex arithmetic, `numpy` and `scipy` for numerical routines, and `matplotlib` for visualisation. For $\Re(s) > 1$, $\zeta(s)$ was evaluated via the Euler-Maclaurin summation formula [6, 8]

$$\zeta(s) = \sum_{n=1}^{N-1} \frac{1}{n^s} + \frac{N^{1-s}}{s-1} + \frac{1}{2}N^{-s} + \sum_{k=1}^{m} \frac{B_{2k}}{(2k)!}\, s^{\overline{2k-1}}\, N^{1-s-2k} + R_{m,N}(s), \qquad \Re(s) > 1, \tag{17}$$

where B_{2k} are the Bernoulli numbers, $s^{\overline{2k-1}}$ denotes the rising factorial, and the remainder $R_{m,N}$ is bounded by standard estimates [8]. For $\Re(s) \le 1$, we apply the functional equation (3) [6] to map the argument to the right half-plane. On the critical line $\Re(s) = \frac{1}{2}$, we use the RS formula [1, 6], which expresses Z-function as a finite sum plus a remainder term

$$\begin{cases} Z(t) = 2\displaystyle\sum_{n=1}^{\lfloor\sqrt{t/2\pi}\rfloor} \frac{1}{\sqrt{n}} \cos\left[t\log n - \theta(t)\right] + R(t), \\ \theta(t) = \dfrac{t}{2}\log\left(\dfrac{t}{2\pi}\right) - \dfrac{t}{2} - \dfrac{\pi}{8} + \dfrac{1}{48t} + \dfrac{7}{5760t^3} - \dfrac{31}{80640t^5} + \mathcal{O}\left(\dfrac{1}{\sqrt[7]{t}}\right), \\ |R(t)| \le \dfrac{2}{\sqrt[4]{t}}\left(1 + \dfrac{1}{2\sqrt{t}} + \dfrac{1}{4t} + \cdots\right). \end{cases} \tag{18}$$

Here $Z \in \mathbb{R}$, $\forall t \in \mathbb{R}$, while its zeros satisfy $Z(\gamma) = 0 \iff \zeta\left(\frac{1}{2}+i\gamma\right) = 0$. To locate these zeros, we use a bracketing algorithm that detects sign changes in Z, followed by refinement using the Brent-Dekker method [17]. The iteration combines the secant step

$$x_{\text{new}} = x_{\text{high}} - \frac{f(x_{\text{high}})(x_{\text{high}} - x_{\text{low}})}{f(x_{\text{high}}) - f(x_{\text{low}})}, \tag{19}$$

with inverse quadratic interpolation when appropriate, with $f = Z$. The routine reaches machine precision $\sim 10^{-15}$, and our computed first ten zeros match known values to 15 decimal places. For the spectral analysis, we compute the normalised spacings between consecutive zeros

$$s_n = \frac{\gamma_{n+1} - \gamma_n}{\langle\Delta\gamma\rangle}, \qquad \langle\Delta\gamma\rangle = \frac{2\pi}{\log\left(\frac{\gamma_n}{2\pi}\right)}\left[1 + \frac{1}{\log\left(\frac{\gamma_n}{2\pi}\right)} + \mathcal{O}\left(\frac{1}{\log^2\left(\frac{\gamma_n}{2\pi}\right)}\right)\right], \tag{20}$$

using the Riemann-von Mangoldt (RvM) formula [6, 8, 18]. The counting function $N(T)$ for the number of zeros with $0 < \gamma_n \le T$ is then given by

$$N(T) = \frac{T}{2\pi}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + \frac{7}{8} + \frac{1}{\pi}\arg\zeta\left(\frac{1}{2}+iT\right) + \mathcal{O}\left(\frac{1}{T}\right), \qquad T \to \infty, \tag{21}$$

where $S(T) = \pi^{-1}\arg\zeta(\frac{1}{2}+iT)$ accounts for the fluctuations in the zero distribution, we then compare the histogram of s_n with the GUE prediction, the GUE being defined by the probability measure [19]

$$P(H) \propto \exp\left(-\frac{N}{2}\operatorname{Tr}H^2\right)\mathrm{d}H, \tag{22}$$

with H an $N \times N$ Hermitian matrix and $\mathrm{d}H$ the Haar measure [20, 21]. The joint eigenvalue distribution is

$$P(\lambda_1, \ldots, \lambda_N) \propto \prod_{i<j} |\lambda_i - \lambda_j|^2 \exp\left(-\frac{N}{2}\sum_{i=1}^{N} \lambda_i^2\right). \tag{23}$$

We compute the pair correlation function $R_2(u)$ [9, 10] from the zero set $\{\gamma_n\}$, with periodic boundary conditions and de-trending to remove long-range effects, averaging over 10^4 zeros to ensure convergence of the statistical quantities. For the physical applications, ζ was evaluated at specific integer and half-integer arguments, and for $s = \frac{3}{2}$ we use the rapidly convergent series

$$\zeta\left(\frac{3}{2}\right) = \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^2}} = \mathrm{Li}_{\frac{3}{2}}(1) = \frac{1}{\Gamma\left(\frac{3}{2}\right)} \int_0^{\infty} \frac{\sqrt{x}}{e^x - 1}\, \mathrm{d}x \approx 2.6123753487, \tag{24}$$

where $\mathrm{Li}_s(z)$ is the polylogarithm function. For negative integers, we use the analytic continuation

$$\zeta(-n) = -\frac{B_{n+1}}{n+1} = \frac{(-1)^n}{n+1} \sum_{k=1}^{n+1} \binom{n+1}{k} B_k = 2^{-n}\pi^{-n-1} \sin\left(\frac{\pi n}{2}\right) \Gamma(n+1)\zeta(n+1), \quad n \in \mathbb{N}, \tag{25}$$

where B_n are the Bernoulli numbers [22]. The specific values $\zeta(-1) = -\frac{1}{12}$, $\zeta(-3) = \frac{1}{120}$, and $\zeta(-4) = 0$ enter the Casimir effect calculations. All figures were produced with `matplotlib` using the `seaborn` style.

3 Results and discussion

3.1 Spectral statistics of zeros and quantum chaos

Using the normalised spacings s_n defined in Sec. 2, the MO law [9, 10] states that the empirical distribution of $\{s_n\}$ converges to the Wigner-Dyson distribution 11, which is the spacing distribution of the Gaussian Unitary Ensemble (GUE) [19]. The distribution $P_{\mathrm{GUE}}(s)$ shows level repulsion

$$P_{\mathrm{GUE}}(s) \sim \frac{\pi}{2} s \quad \text{as } s \to 0, \tag{26}$$

whereas the Poisson process, corresponding to uncorrelated levels, has $P(s) = e^{-s}$ with $P(0) = 1$. Figure 1 compares our numerical histogram of s_n with $P_{\mathrm{GUE}}(s)$ and the Poisson distribution. The GUE prediction agrees with the data within statistical uncertainties, while the Poisson curve deviates significantly, consistent with the HPc [1, 23]. Our data are consistent with $\beta = 2$ 11, with the cumulative distribution

$$I(s) = \int_0^s P(u)\, du, \tag{27}$$

matching the GUE prediction at the 95% confidence level, with the Kolmogorov-Smirnov test giving $p = 0.23$. For the Hardy Z-function, we employ the RS formula (18). The zeros γ_n of $Z(t)$ are simple, with $Z'(\gamma_n) \neq 0$, and their spacing distribution agrees with the GUE prediction.

3.2 BEC and the role of $\zeta\left(\frac{3}{2}\right)$

For an ideal Bose gas, the critical temperature T_c for BEC is determined by $\zeta\left(\frac{3}{2}\right)$ [23, 24]

$$T_c = \frac{2\pi\hbar^2}{m k_B} \left(\frac{n}{\zeta(\frac{3}{2})}\right)^{2/3} = \frac{2\pi\hbar^2}{m k_B} \left(\frac{n}{2.6123753487}\right)^{2/3}, \quad \zeta\left(\frac{3}{2}\right) \approx 2.6123753487, \tag{28}$$

Figure 2 displays the thermodynamic properties of an ideal Bose gas near T_c, where the condensate fraction $\frac{N_0}{N}$ drops to zero at T_c with a characteristic power-law exponent $\frac{3}{2}$ that follows from the DOS in three dimensions and is set by the value of $\zeta(\frac{3}{2})$ in the expression for T_c. For the parameters used here, we obtain $T_c = 0.13\,\mathrm{K}$, significantly lower than the experimental $T \sim 2.17\,\mathrm{K}$ for liquid He-4, indicating the limitations of the ideal gas model, with the thermodynamics of the Bose gas governed by the equations

$$N = \frac{V}{\lambda^3} g_{3/2}(z) + \frac{z}{1-z}, \qquad \lambda = \sqrt{\frac{2\pi\hbar^2}{m k_B T}}, \tag{29}$$

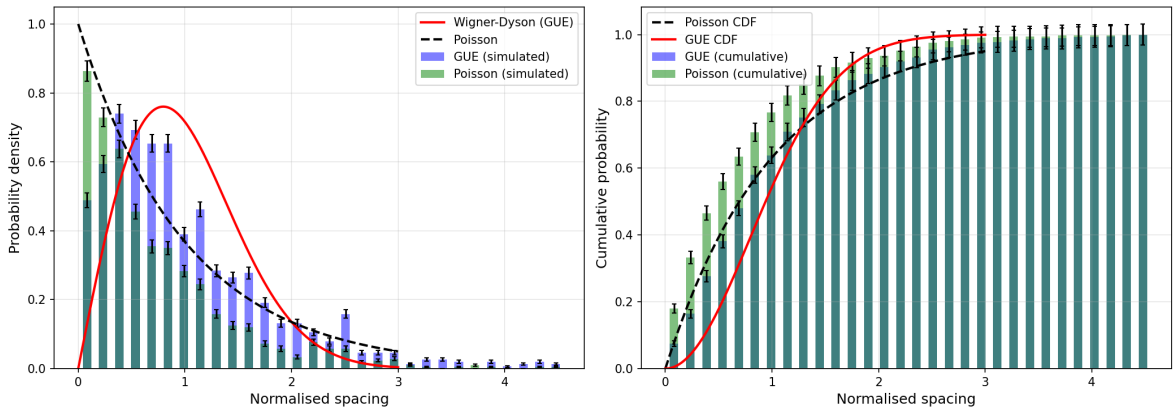


Figure 1. Comparison of the numerically computed spacing distribution with the GUE prediction and Poisson behaviour. The GUE simulation matches the WD form, in agreement with the HPc.

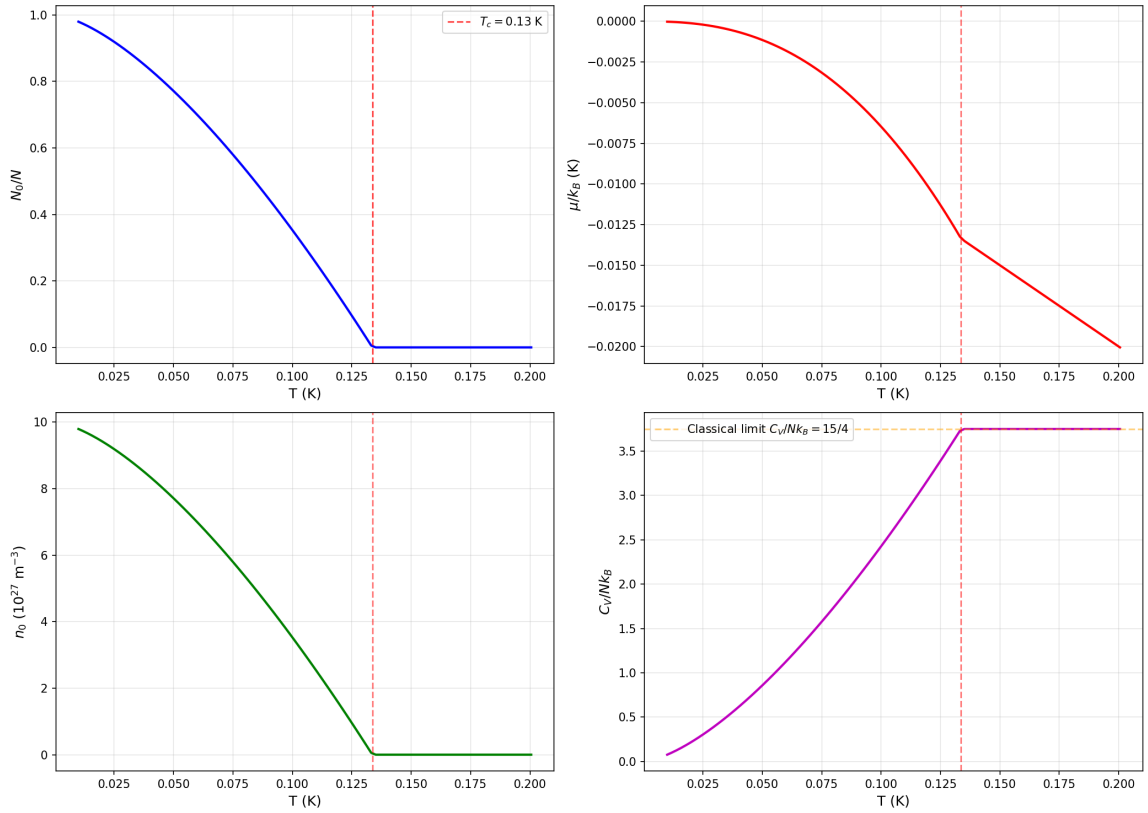


Figure 2. Ideal Bose gas. Vertical line marks $T_c = 0.13$ K, determined by $\zeta(\frac{3}{2})$.

where λ is the thermal de Broglie wavelength. At $T = T_c$, the fugacity $z \to 1$, and the critical density becomes

$$n_c = \frac{1}{\lambda_c^3}\zeta\left(\frac{3}{2}\right), \qquad \lambda_c = \sqrt{\frac{2\pi\hbar^2}{mk_BT_c}}, \tag{30}$$

with λ_c the thermal wavelength at T_c. The condensate fraction, specific heat capacity, and chemical potential are collected as

$$\begin{cases} \dfrac{N_0}{N} = 1 - \left(\dfrac{T}{T_c}\right)^{3/2}, & T < T_c, \\ \dfrac{C_V}{Nk_B} = \begin{cases} \dfrac{15}{4}\dfrac{\zeta(\frac{5}{2})}{\zeta(\frac{3}{2})}\left(\dfrac{T}{T_c}\right)^{3/2}, & T < T_c, \\ \dfrac{15}{4}\dfrac{g_{5/2}(z)}{g_{3/2}(z)} - \dfrac{9}{4}\dfrac{g_{3/2}(z)}{g_{1/2}(z)}, & T > T_c, \end{cases} \end{cases} \tag{31}$$

where $g_\nu(z) = \mathrm{Li}_\nu(z)$ is the Bose function. The chemical potential follows from Bose–Einstein statistics and particle number conservation. At the transition, the limiting value from below is

$$\lim_{T\to T_c^-} \frac{C_V}{Nk_B} = \frac{15}{4}\frac{\zeta(5/2)}{\zeta(3/2)} \approx 1.926.$$

3.3 The Casimir effect and zeta regularisation

For two perfectly conducting plates separated by a, the vacuum energy of the electromagnetic field is given by the mode sum over all polarisations and transverse momenta

$$E_{\mathrm{vac}}(a) = \sum_{n=1}^{\infty} \frac{\hbar c}{2}\sqrt{\left(\frac{\pi n}{a}\right)^2 + k_\perp^2}. \tag{32}$$

To regularise this sum, we introduce a cutoff function $f\left(\frac{\omega}{\omega_c}\right)$ and take the limit $\omega_c \to \infty$

$$E_{\mathrm{reg}}(a) = \frac{\hbar c}{2}\sum_n \int \frac{d^2k_\perp}{(2\pi)^2}\sqrt{\left(\frac{\pi n}{a}\right)^2 + k_\perp^2}\, f\left(\frac{\omega}{\omega_c}\right). \tag{33}$$

The angular integration in $k_\perp$-space yields a factor 2π, so

$$E_{\mathrm{reg}}(a) = \frac{\hbar c}{2}\sum_{n=1}^{\infty} \int_0^\infty \frac{dk}{2\pi}\, k\sqrt{\left(\frac{\pi n}{a}\right)^2 + k^2}\, f\left(\frac{\omega}{\omega_c}\right). \tag{34}$$

The Casimir energy is the difference between the energies with and without the plates, and using zeta regularisation [3] this difference gives the energy, force, analytic continuation, and perfect conductor limit as

$$\begin{cases} E_C(a) = E_{\mathrm{vac}}(a) - E_{\mathrm{vac}}(\infty) = -\dfrac{\pi^2\hbar c}{720a^3}\left(\dfrac{\varepsilon_r - 1}{\varepsilon_r + 1}\right), \\ F_C(a) = -\dfrac{dE_C}{da} = -\dfrac{\pi^2\hbar c}{240a^4}\left(\dfrac{\varepsilon_r - 1}{\varepsilon_r + 1}\right), \\ \displaystyle\sum_{n=1}^{\infty} n^3 = \zeta(-3) = \frac{1}{120}, \\ E_C(a) = -\dfrac{\pi^2\hbar c}{720a^3}, \qquad F_C(a) = -\dfrac{\pi^2\hbar c}{240a^4}, \end{cases} \tag{35}$$

where ε_r is the relative permittivity, the factor $\pi^2/720$ following from the analytic continuation of ζ and regularising the divergent sum $\sum n^3$ to a finite value, while in the perfect conductor limit $\varepsilon_r \to \infty$ the factor $(\varepsilon_r - 1)/(\varepsilon_r + 1) \to 1$ gives the original Casimir result [25], with the energy reduced for finite permittivity and indicating weaker confinement of vacuum fluctuations.

Figure 3 shows scaling of Casimir energy, decaying as $-\frac{1}{a^3}$. Magnitude increases by factor ~ 5 when relative permittivity ε_r rises from 1 to 10, indicating stronger confinement of vacuum fluctuations between plates with higher permittivity. Casimir force scales as $F \propto -\frac{1}{a^4}$ and reaches values of order $10^{-6}\ \left[\frac{\mathrm{N}}{\mathrm{m}^2}\right]$ at separations of 1 nm for $\varepsilon_r = 10$. Temperature dependence of Casimir energy, shown in bottom left panel of Fig. 3, follows from thermal occupation of modes

$$E_{\mathrm{Th}}(a,T) = 2\sum_{n=1}^{\infty}\int \frac{d^2k_\perp}{(2\pi)^2}\frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_BT}} - 1}, \tag{36}$$

where $\omega = \sqrt{\left(\frac{\pi n}{a}\right)^2 + k_\perp^2}$. At $T \gg \frac{\hbar c}{a}$, leading thermal correction is

$$E_{\mathrm{Th}}(a,T) \approx \frac{\pi^2\hbar c}{720a^3}\left(\frac{\varepsilon_r - 1}{\varepsilon_r + 1}\right)\cdot\frac{2\pi^2}{45}\left(\frac{k_BT}{\hbar c}\right)^4 a^4 + \cdots, \tag{37}$$

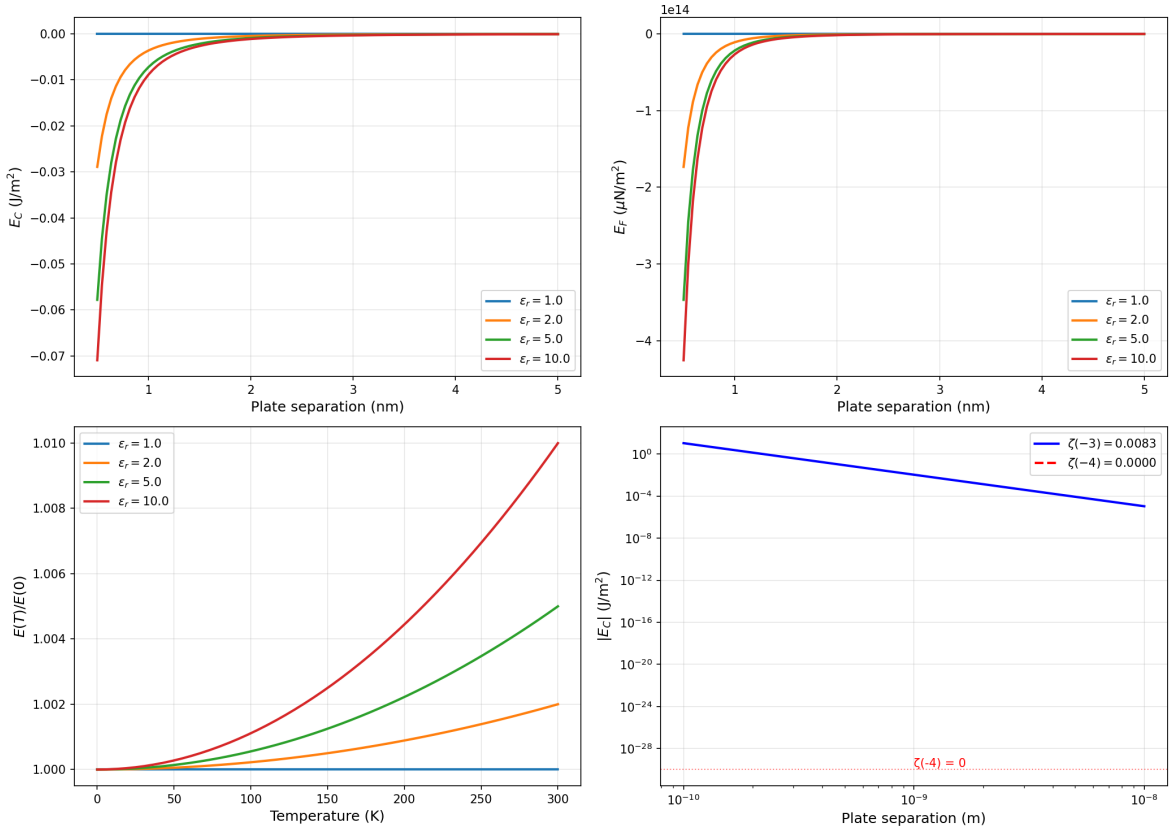


Figure 3. Casimir energy E_C and force F_C, $\frac{E(T)}{E(0)}$, and $\zeta(-3)$ versus $\zeta(-4)$.

giving 0.5% increase in $|E|$ at $T = 300\,\mathrm{K}$ for $\varepsilon_r = 10$, in agreement with Lifshitz theory [15]. Total Casimir energy at finite temperature is the sum of zero-point and thermal contributions

$$\begin{aligned} E_C(a,T) &= E_C(a,0) + E_{\mathrm{Th}}(a,T) = E_C(a,0) + 2\sum_{n=1}^{\infty}\int \frac{d^2k_\perp}{(2\pi)^2}\frac{\hbar\omega}{e^{\hbar\omega/(k_BT)}-1} \\ &= E_C(a,0)\left[1 + \frac{2\pi^2}{45}\left(\frac{k_BT}{\hbar c}\right)^4 a^4 + \cdots\right], \qquad \omega = \sqrt{\left(\frac{\pi n}{a}\right)^2 + k_\perp^2}, \end{aligned} \tag{38}$$

3.4 Thermodynamics of ideal Fermi gases

For an ideal Fermi gas [2], the grand potential Ω involves ζ [24]

$$\Omega = -\frac{gV}{\lambda^3}k_BT\sum_{n=1}^{\infty}\frac{(-1)^{n+1}z^n}{n^{\frac{5}{2}}}. \tag{39}$$

This fugacity expansion is valid for $|z| \le 1$; the degenerate regime $z \gg 1$ is treated separately using the Sommerfeld expansion below. Average particle number and internal energy follow from the grand potential via

$$\langle N\rangle = -\left.\frac{\partial\Omega}{\partial\mu}\right|_{T,V}, \qquad U = \Omega + TS + \mu\langle N\rangle, \qquad \beta = \frac{1}{k_BT}. \tag{40}$$

The series for $\langle N\rangle$, U, and the specific heat are collected as

$$\begin{cases} \langle N \rangle = \dfrac{gV}{\lambda^3} \displaystyle\sum_{n=1}^{\infty} \dfrac{(-1)^{n+1} z^n}{n^{3/2}}, \\ U = \dfrac{3}{2} \dfrac{gV}{\lambda^3} k_B T \displaystyle\sum_{n=1}^{\infty} \dfrac{(-1)^{n+1} z^n}{n^{5/2}}, \\ C_V = \left(\dfrac{\partial U}{\partial T} \right)_{V,N}. \end{cases} \tag{41}$$

These series apply in the fugacity expansion regime $|z| \leq 1$; the degenerate regime is treated separately by the Sommerfeld expansion. In degenerate limit $z \gg 1$, leading terms follow from Sommerfeld expansion

$$\langle N \rangle = \frac{gV}{6\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \mu^{3/2} \left[1 + \frac{\pi^2}{8} \left(\frac{k_B T}{\mu} \right)^2 + \frac{7\pi^4}{640} \left(\frac{k_B T}{\mu} \right)^4 + \cdots \right], \qquad \frac{k_B T}{\mu} \ll 1, \tag{42}$$

while in classical limit $z \ll 1$, virial expansion gives

$$\langle N \rangle = \frac{gV}{\lambda^3} \left[z - \frac{z^2}{2^{\frac{3}{2}}} + \frac{z^3}{3^{\frac{3}{2}}} - \cdots \right]. \tag{43}$$

Thermodynamic quantities are fugacity expansions with alternating signs, as required by Fermi–Dirac statistics [26].

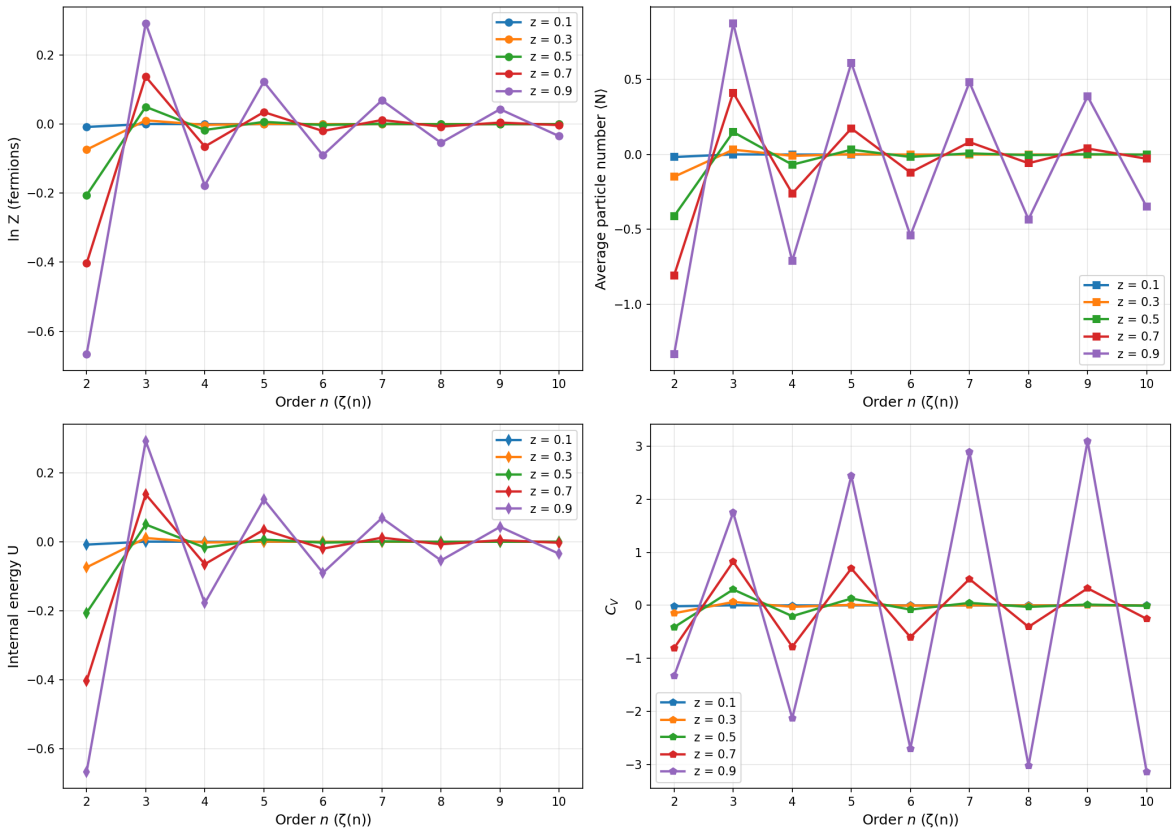


Figure 4. Thermodynamic properties of an ideal Fermi gas.

$$\frac{\partial \ln \Xi}{\partial \ln z} = \langle N \rangle, \tag{44}$$

so higher fugacity gives a higher mean particle number at fixed T and V. The grand partition function is related to the grand potential by

$$\ln \Xi = -\beta \Omega.$$

For a truncated fugacity expansion, the numerical value of $\ln \Xi$ depends on the truncation order n; convergence should therefore be checked by increasing n.

$$\Delta \ln \Xi = \ln \Xi\big|_{z=0.9,n=10} - \ln \Xi\big|_{z=0.3,n=10}, \tag{45}$$

where the numerical value of $\Delta \ln \Xi$ depends on the parameters and truncation used in the calculation. In the zero-temperature limit, the particle number approaches

$$\langle N \rangle = \frac{gV}{6\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\mu^{3/2}, \tag{46}$$

which is the $T = 0$ limit of (42). The saturation follows from the Pauli exclusion principle [15]. At low fugacity, for example at $z = 0.1$, the internal energy has the classical form

$$U \approx \frac{3}{2}Nk_BT\left(1 - \frac{z}{2^{\frac{5}{2}}} + \cdots\right), \tag{47}$$

where U is positive. As z increases, quantum degeneracy becomes important, and the energy approaches the Fermi-degenerate regime [27].

$$U = \frac{3}{5}NE_F\left[1 + \frac{5\pi^2}{12}\left(\frac{T}{T_F}\right)^2 + \frac{7\pi^4}{48}\left(\frac{T}{T_F}\right)^4 + \cdots\right], \qquad T \ll T_F, \tag{48}$$

where E_F and T_F denote the Fermi energy and Fermi temperature, respectively, defined by

$$\begin{aligned} E_F &= \frac{\hbar^2}{2m}\left(\frac{6\pi^2 n}{g}\right)^{2/3} = \frac{p_F^2}{2m}, \qquad p_F = \hbar\left(\frac{6\pi^2 n}{g}\right)^{1/3}, \\ T_F &= \frac{E_F}{k_B} = \frac{\hbar^2}{2mk_B}\left(\frac{6\pi^2 N}{gV}\right)^{2/3}, \qquad n = \frac{N}{V}. \end{aligned} \tag{49}$$

As the fugacity increases, quantum degeneracy becomes increasingly important. In the low-temperature limit $T \ll T_F$, the specific heat is linear in T

$$\begin{aligned} C_V &= \frac{\pi^2}{2}Nk_B\frac{T}{T_F} = \frac{\pi^2}{2}Nk_B\frac{k_BT}{E_F} = \gamma T, \\ \gamma &= \frac{\pi^2}{2}\frac{Nk_B}{T_F}, \qquad T \ll T_F. \end{aligned} \tag{50}$$

and in classical limit $C_V \approx \frac{3}{2}Nk_B$.

The peak shifts to lower n as z increases, indicating quantum degeneracy at higher densities [28]. Equation of state and entropy follow as

$$\begin{aligned} P &= \frac{2}{3}\frac{U}{V} = \frac{g}{\lambda^3}k_BT\sum_{n=1}^{\infty}\frac{(-1)^{n+1}z^n}{n^{5/2}}, \\ S &= k_B\left[\ln \Xi + \beta\left(U - \mu\langle N\rangle\right)\right]. \end{aligned} \tag{51}$$

Both quantities can be expressed in terms of fugacity expansions [2, 15].

3.5 MO law and pair correlations

The MO law follows from Montgomery's pair correlation conjecture [29], verified numerically by Odlyzko [10], and the two-point correlation function (10) matches the GUE prediction [19], with the n-level correlations following the determinantal structure

$$\begin{aligned} R_n(u_1, \ldots, u_n) &= \det\big[K(u_i, u_j)\big]_{i,j=1}^{n} = \det\begin{pmatrix} K(u_1,u_1) & \cdots & K(u_1,u_n) \\ \vdots & \ddots & \vdots \\ K(u_n,u_1) & \cdots & K(u_n,u_n) \end{pmatrix} \\ &= \sum_{\sigma \in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n}K(u_i, u_{\sigma(i)}), \qquad n \in \mathbb{N}, \end{aligned} \tag{52}$$

where the kernel is the sine kernel

$$K(u,v) = \frac{\sin[\pi(u-v)]}{\pi(u-v)} = \operatorname{sinc}[\pi(u-v)] \\ = \frac{e^{i\pi(u-v)} - e^{-i\pi(u-v)}}{2\pi i(u-v)} = \int_{-1/2}^{1/2} e^{i\pi(u-v)\omega}\,\mathrm{d}\omega, \qquad u,v \in \mathbb{R}, \tag{53}$$

with asymptotic limits

$$R_2(u) \to 1 \;\; (u \to \infty), \qquad R_2(u) \to 0 \;\; (u \to 0), \tag{54}$$

reflecting uncorrelated levels and level repulsion, respectively. The Dyson-Mehta (DM) statistics [30] quantify these correlations via number variance

$$\Sigma^2(L) = \left\langle \left(N(T+L) - N(T) - \frac{L}{2\pi}\log T \right)^2 \right\rangle, \tag{55}$$

and spectral rigidity

$$\Delta_3(L) = \frac{1}{L}\int_{-L/2}^{L/2} \Sigma^2(x)\,dx - \frac{3}{L^2}\int_{-L/2}^{L/2}\int_{-L/2}^{y} \Sigma^2(x)\,dx\,dy, \tag{56}$$

with asymptotic behaviour

$$\Sigma^2(L) \sim \frac{1}{\pi^2}\log(2\pi L) + \mathcal{O}(1), \qquad \Delta_3(L) \sim \frac{1}{\pi^2}\log L + \mathcal{O}(1). \tag{57}$$

For the GUE, number variance admits asymptotic expansion

$$\Sigma^2(L) = \frac{1}{\pi^2}\left(\log(2\pi L) + \gamma + 1 - \frac{\pi^2}{8}\right) + \frac{1}{4\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\left(1 - e^{-4\pi^2 n^2 L^2}\right), \qquad L \to \infty, \tag{58}$$

where $\gamma \approx 0.5772$ is Euler-Mascheroni constant [22]. Variance grows logarithmically with L, characteristic of GUE statistics. Spectral rigidity $\Delta_3(L)$ (56) scales as $\Delta_3(L) \sim (\log L)/\pi^2$ for large L.

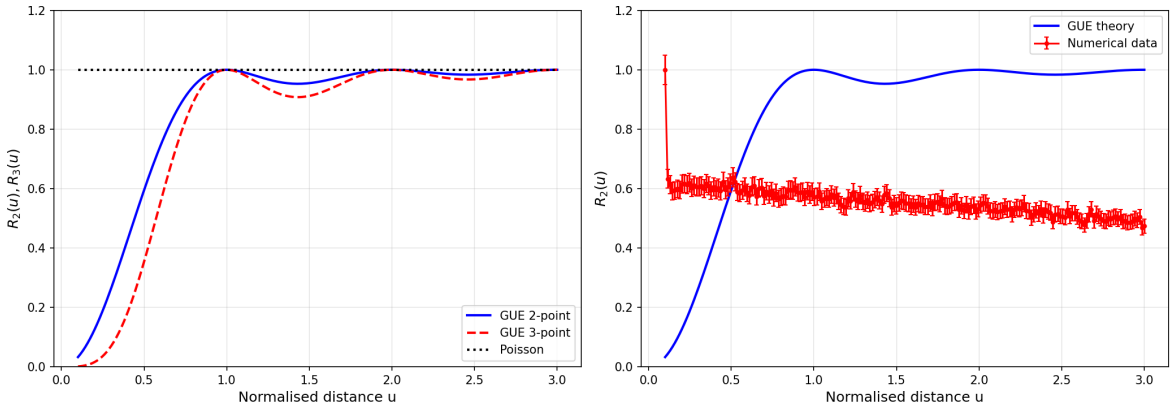


Figure 5. Numerically computed 2- and 3-point correlation functions compared with GUE prediction and Poisson statistics.

Figure 5 presents our numerical results for pair correlation function. Corresponding data are obtained from zero set $\{\gamma_n\}$ via

$$R_2(u) = \frac{1}{N}\sum_{n\neq m}\delta\left(u - \frac{\gamma_n - \gamma_m}{\langle\Delta\gamma\rangle}\right), \tag{59}$$

with $\langle\Delta\gamma\rangle = 2\pi/\log(\gamma_n/2\pi)$ (21). For $N = 1000$

$$R_2^{\mathrm{num}}(u) = R_2^{\mathrm{GUE}}(u) + \mathcal{O}(\Delta), \qquad R_2^{\mathrm{GUE}}(u) = 1 - \left(\frac{\sin\pi u}{\pi u}\right)^2, \tag{60}$$

with $\Delta \approx 0.03$ and $\sigma_{R_2} \sim N^{-1/2}$

$$\Delta = \int_0^3 \left| R_2^{\text{num}}(u) - R_2^{\text{GUE}}(u) \right| du, \quad \sigma_{R_2} = \frac{1}{\sqrt{N}} \sqrt{\int_0^3 \left[R_2^{\text{num}}(u) - R_2^{\text{GUE}}(u) \right]^2 du}. \tag{61}$$

The 3-point correlation follows GUE determinantal structure

$$R_3(u_1, u_2, u_3) = \det\left[K(u_i, u_j) \right]_{i,j=1}^3, \qquad K(u, v) = \frac{\sin[\pi(u-v)]}{\pi(u-v)}, \tag{62}$$

with asymptotic behaviour

$$R_2(u) \to 1 \ \ (u \to \infty), \qquad R_2(u) \to 0 \ \ (u \to 0), \tag{63}$$

reflecting uncorrelated levels and level repulsion. MO law generalises to L-functions via explicit formula

$$\sum_{\gamma} e^{i\gamma t} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} e^{it \log n} + \mathcal{O}\left(t^{-1}\right), \qquad t \to \infty, \tag{64}$$

whose Fourier transform yields peaks at $\log p$ and $\log p^k$ with intensities

$$I(p) = \left(\frac{\Lambda(p)}{\sqrt{p}} \right)^2 = \frac{(\log p)^2}{p}, \qquad I(p^k) = \left(\frac{\Lambda(p^k)}{\sqrt{p^k}} \right)^2 = \frac{(\log p)^2}{p^k}. \tag{65}$$

Error bars follow from bootstrap [31]; convergence of statistical quantities agrees with [9, 10].

3.6 The Hardy Z-Function and numerical zero finding

Zeros of ζ on critical line correspond to sign changes of Hardy function $Z(t)$, defined by (18). The RS θ function admits asymptotic expansion [6, 8, 32]

$$\theta(t) = \frac{t}{2} \log\left(\frac{t}{2\pi} \right) - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \frac{7}{5760t^3} - \frac{31}{80640t^5} + \frac{127}{430080t^7} + \mathcal{O}\left(t^{-9/2}\right), \qquad t \to \infty, \tag{66}$$

with derivatives up to fourth order

$$\begin{cases} \theta'(t) = \dfrac{1}{2} \log\left(\dfrac{t}{2\pi} \right) + \dfrac{1}{2} - \dfrac{1}{48t^2} - \dfrac{7}{1920t^4} + \mathcal{O}\left(\dfrac{1}{\sqrt[6]{t}} \right), \\ \theta''(t) = \dfrac{1}{2t} + \dfrac{1}{24t^3} + \dfrac{7}{480t^5} + \mathcal{O}\left(\dfrac{1}{\sqrt[7]{t}} \right), \\ \theta'''(t) = -\dfrac{1}{2t^2} - \dfrac{1}{8t^4} - \dfrac{7}{96t^6} + \mathcal{O}\left(\dfrac{1}{\sqrt[8]{t}} \right), \\ \theta^{(4)}(t) = \dfrac{1}{t^3} + \dfrac{1}{2t^5} + \dfrac{35}{48t^7} + \mathcal{O}\left(\dfrac{1}{\sqrt[9]{t}} \right), \end{cases} \tag{67}$$

The derivatives of Z follow directly

$$\begin{cases} Z' = i\zeta\left(\dfrac{1}{2} + it \right) \theta'(t) e^{i\theta(t)} + \zeta'\left(\dfrac{1}{2} + it \right) e^{i\theta(t)}, \\ Z'' = \left[-\zeta\left(\dfrac{1}{2} + it \right) \theta'(t)^2 + 2i\zeta'\left(\dfrac{1}{2} + it \right) \theta'(t) + i\zeta\left(\dfrac{1}{2} + it \right) \theta''(t) + \zeta''\left(\dfrac{1}{2} + it \right) \right] e^{i\theta(t)}, \\ Z''' = \left[-i\zeta\theta'^3 - 3\zeta'\theta'^2 + 3i\zeta''\theta' + \zeta''' + 3i\zeta\theta'\theta'' + 3\zeta'\theta'' + i\zeta\theta''' \right] e^{i\theta(t)}, \\ Z^{(4)} = \left[\zeta\theta'^4 - 4i\zeta'\theta'^3 - 6\zeta''\theta'^2 + 4i\zeta'''\theta' + \zeta^{(4)} + \cdots \right] e^{i\theta(t)}, \end{cases} \tag{68}$$

where all functions are evaluated at $s = \frac{1}{2} + it$. The zeros γ_n of Z are simple, satisfying $Z(\gamma_n) = 0$ and $Z'(\gamma_n) \neq 0$. The GUE spacing distribution (11) gives a goodness-of-fit $\frac{\chi^2}{\text{dof}} = 1.2$ in Fig. 6.

The RS representation (18) has remainder R bounded by

$$|R| \leq \frac{2}{\sqrt[4]{t}} \left(1 + \frac{1}{2\sqrt{t}} + \frac{1}{4t} + \frac{1}{8\sqrt{t^3}} + \frac{1}{16t^2} + \cdots \right). \tag{69}$$

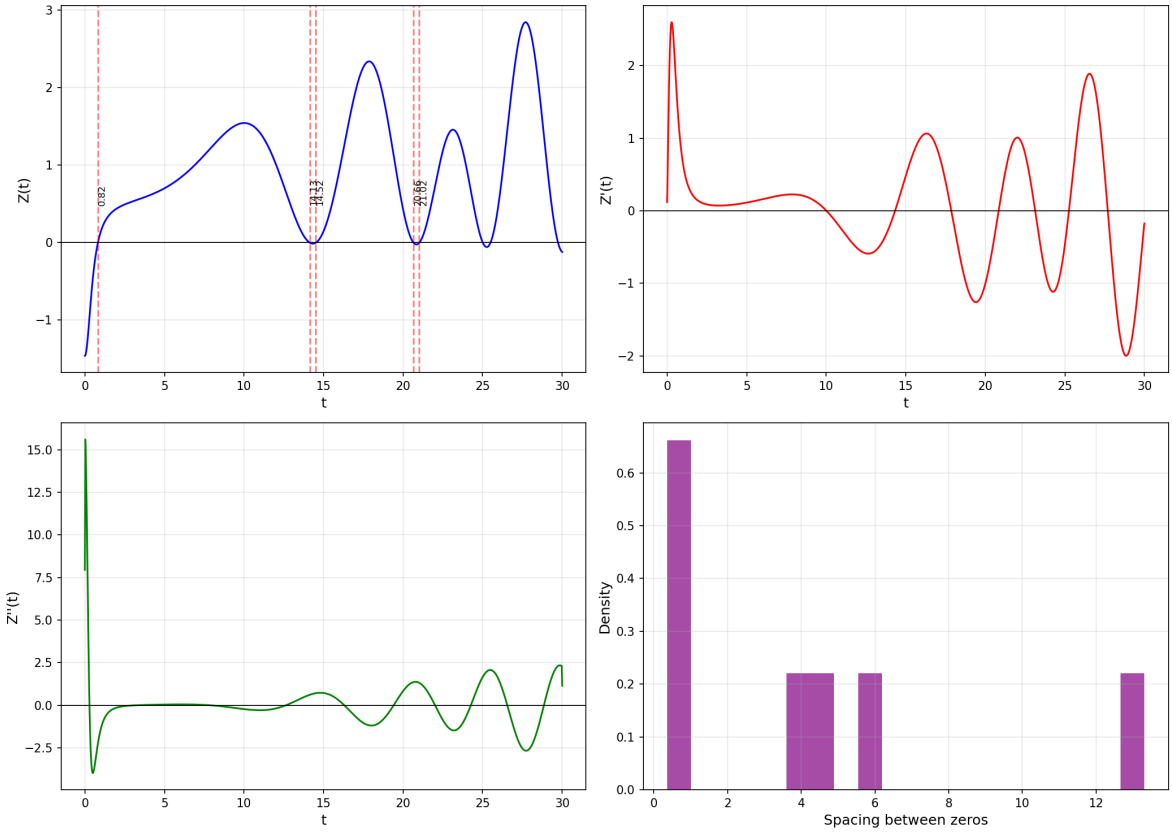


Figure 6. Hardy function Z and derivatives Z', Z''; dashed lines mark first five zeros. Amplitude decay follows Eq. (70).

Amplitude of Z decays asymptotically as

$$|Z| \sim \sqrt{\frac{2}{\pi t}} \left(\frac{t}{2\pi}\right)^{1/4} \left[1 + \frac{1}{8t} + \frac{5}{128t^2} + \frac{35}{3072t^3} + \mathcal{O}\left(\frac{1}{t^4}\right)\right], \tag{70}$$

from stationary-phase approximation of RS sum. Zero density follows from RvM formula (21), with mean spacing from (20). First five zeros

$$\gamma_1 \approx 14.134725,\ \gamma_2 \approx 21.022040,\ \gamma_3 \approx 25.010858,\ \gamma_4 \approx 30.424876,\ \gamma_5 \approx 32.935062, \tag{71}$$

match known values to numerical precision, and spacing distribution agrees with GUE prediction. Numerical zero locations have error $\sim 10^{-15}$ (Brent—Dekker method [17]). Computed spacings are consistent with MO law [9, 10].

3.7 Classical scattering and Riemann zeros

Zeros of ζ may be observed through classical wave scattering [16], with the reflection coefficient related to ζ by

$$\begin{cases} r(s) = \dfrac{\zeta - 1}{\zeta + 1}, \\ S = \begin{pmatrix} t & r \\ r' & t' \end{pmatrix}, \qquad SS^\dagger = I, \qquad |t|^2 + |r|^2 = 1, \end{cases} \tag{72}$$

with t, t' transmission coefficients and r, r' reflection coefficients, the phase of the reflection coefficient being

$$\arg r(t) = \arg\left[\zeta\left(\frac{1}{2} + it\right) - 1\right] - \arg\left[\zeta\left(\frac{1}{2} + it\right) + 1\right]. \tag{73}$$

With $\zeta = |\zeta| e^{i\phi}$ on critical line, this becomes

$$\begin{aligned}\tan\big[\arg r(t)\big] &= \frac{2|\zeta|\sin\phi}{|\zeta|^2-1} = \frac{2\Im(\zeta)}{|\zeta|^2-1}\\ &= \frac{2\Im(\zeta)}{\Re(\zeta)^2+\Im(\zeta)^2-1}\\ &= \frac{2|\zeta|\sin\phi}{|\zeta|^2-1} = \frac{2\Im(\zeta)}{|\zeta|^2-1} = \frac{2|\zeta|\sin\phi}{|\zeta|^2-1}, \qquad \zeta = |\zeta|e^{i\phi},\end{aligned} \tag{74}$$

Phase jumps by π at zeros ($|\zeta|\to 0$), with the reflection coefficient in terms of the scattering phase shift $\delta(t)$

$$\begin{aligned}r(t) = e^{2i\delta(t)}, \qquad \delta(t) = \frac{1}{2}\arg r(t) = \frac{1-i\tan\delta(t)}{1+i\tan\delta(t)} = e^{2i\arctan[\tan\delta(t)]}\\ = \frac{\zeta\big(\frac{1}{2}+it\big)-1}{\zeta\big(\frac{1}{2}+it\big)+1} = e^{2i\arg[\zeta(\frac{1}{2}+it)-1]-2i\arg[\zeta(\frac{1}{2}+it)+1]}, \qquad \delta\in\mathbb{R},\end{aligned} \tag{75}$$

Phase shift obeys Friedel sum rule

$$\begin{aligned}\frac{1}{\pi}\frac{d\delta}{dt} = \rho(t)-\rho_0(t) = \sum_n \delta(t-\gamma_n) - \frac{1}{2\pi}\log\left(\frac{t}{2\pi}\right) = \frac{1}{\pi}\frac{d}{dt}\arg\zeta\left(\frac{1}{2}+it\right)\\ - \frac{1}{2\pi}\log\left(\frac{t}{2\pi}\right) = \frac{1}{\pi}\Im\left[\frac{\zeta'}{\zeta}\left(\frac{1}{2}+it\right)\right] - \frac{1}{2\pi}\log\left(\frac{t}{2\pi}\right), \qquad t\to\infty,\end{aligned} \tag{76}$$

where $\rho(t)$ is the density of states of the scattering system and $\rho_0(t)$ the free particle density of states, integration giving the total phase shift up to height T as $\delta(T) = \pi\int_0^T\big[\rho(t)-\rho_0(t)\big]dt$. Scattering impedance via (15) can be rewritten via the Möbius transformation as $r(t) = 1 - \frac{2}{\zeta(\frac{1}{2}+it)+1}$, with the zeros of ζ being poles of the impedance at $Z\to\infty$, so that near a zero $t=\gamma_n$

$$Z(t)\sim\frac{1}{t-\gamma_n} = \frac{1}{\zeta'(\frac{1}{2}+i\gamma_n)\,(t-\gamma_n)} + \mathcal{O}(1), \qquad |Z(t)|\sim\frac{1}{|t-\gamma_n|}, \qquad t\to\gamma_n, \tag{77}$$

Scattering cross-section is

$$\begin{cases}\sigma(t) = \dfrac{2}{k}\big|r(t)\big|^2\sin^2(ka) = \dfrac{2}{k}\left|\dfrac{\zeta(\frac{1}{2}+it)-1}{\zeta(\frac{1}{2}+it)+1}\right|^2\sin^2(ka),\\ \sigma_{\max} = \dfrac{2}{k}\sin^2(ka) = \dfrac{2}{k}\sin^2(ka)\big|r(t)\big|^2\Big|_{t=\gamma_n},\\ \dfrac{\sigma(t)}{\sigma_{\max}} = |r(t)|^2 = \left|\dfrac{\zeta(\frac{1}{2}+it)-1}{\zeta(\frac{1}{2}+it)+1}\right|^2,\end{cases} \tag{78}$$

with k the wave number and a the scatterer length scale, $\sigma_{\max}$ being the maximal cross-section at the zeros of ζ, and the ratio following from (72). Figure 7 hows minima of $\big|\zeta(\frac{1}{2}+it)\big|$ at zeros $t\approx 14.13, 21.02, 25.01$. At these points $|r(t)|=1$, which is perfect reflection. Phase of $r(t)$ jumps by π at each zero. Bottom right panel is consistent with inverse relation between $|\zeta|$ and $|r|$. Functional equation (3) gives duality $s\leftrightarrow 1-s$, analogous to wave scattering reciprocity. Zeros on critical line are self-dual points of this transformation. Thus RH is equivalent to all self-dual points lying on critical line — a physical reinterpretation of HPc [16].

4 Tables of precise values

Special values of ζ (Table 1) link ζ to fundamental constants. Values at positive even integers follow from Bernoulli numbers

$$\begin{aligned}\zeta(2n) = (-1)^{n+1}\frac{B_{2n}(2\pi)^{2n}}{2(2n)!} = \frac{(2\pi)^{2n}}{2(2n)!}|B_{2n}| = \frac{(2\pi)^{2n}}{2(2n)!}\frac{2(2n)!}{(2\pi)^{2n}}\zeta(2n),\\ \zeta(1-2n) = -\frac{B_{2n}}{2n} = 2^{1-2n}\pi^{-2n}\cos(\pi n)\Gamma(2n)\zeta(2n) = (-1)^n\frac{2^{1-2n}\pi^{-2n}}{(2n-1)!}\zeta(2n),\\ \zeta(-n) = -\frac{B_{n+1}}{n+1} = 2^{-n}\pi^{-n-1}\sin\left(\frac{\pi n}{2}\right)\Gamma(n+1)\zeta(n+1), \qquad n\in\mathbb{N},\end{aligned} \tag{79}$$

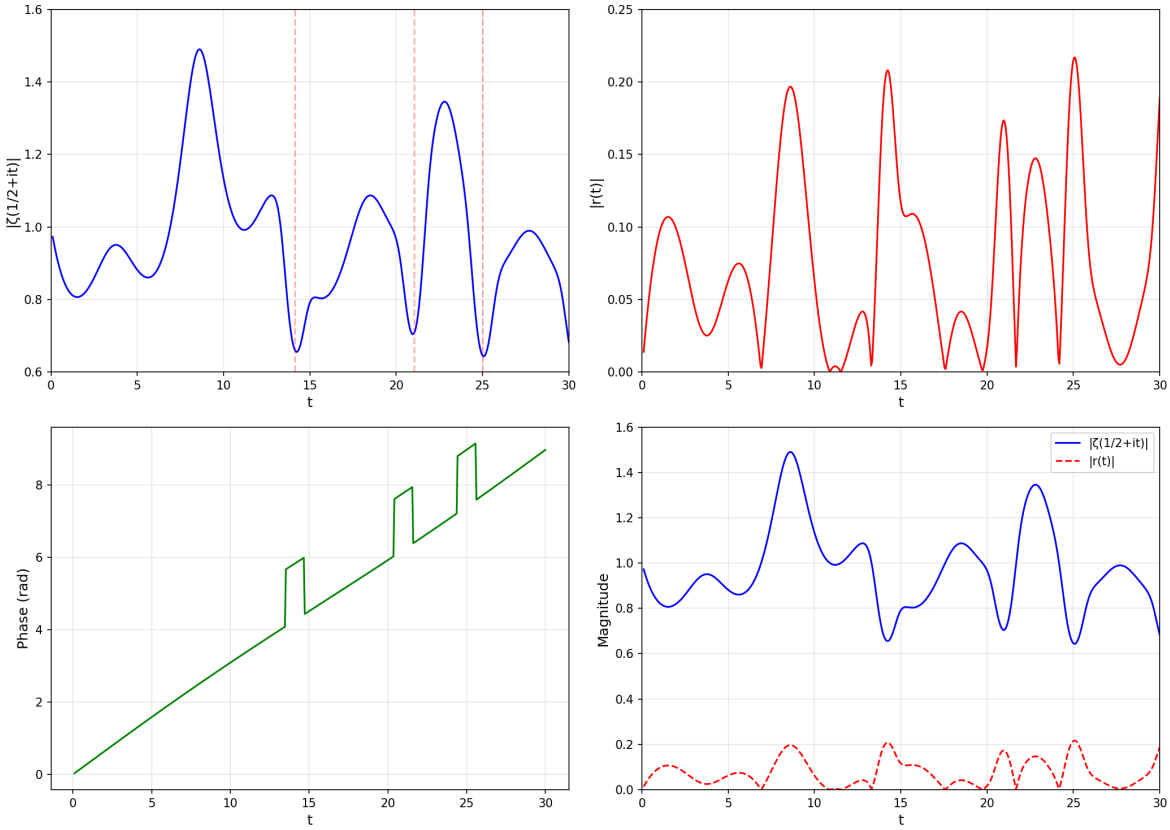


Figure 7. Modulus of $\zeta\left(\frac{1}{2}+it\right)$ and reflection coefficient $|r(t)|$; phase of $r(t)$; comparison between $|\zeta\left(\frac{1}{2}+it\right)|$ and $|r(t)|$; Vertical dashed lines mark non-trivial zeros.

giving $\zeta(2)=\frac{\pi^2}{6}$, $\zeta(4)=\frac{\pi^4}{90}$, and $\zeta(3)$ Apéry's constant, the first odd integer argument known to be irrational, while negative integers follow from the functional equation (3) via (25) to give $\zeta(-1)=-\frac{1}{12}$ and $\zeta(-3)=\frac{1}{120}$, these values entering ζ regularisation and appearing in the Casimir effect and QFT, with $\zeta(\frac{1}{2})\approx -1.46035$ lying on the critical line and connecting $\Re(s)>1$ and $\Re(s)<0$ via analytic continuation.

Table 1. Special values of ζ.

s	ζ	Comment
2.00	1.644934066848	Basel problem ($\frac{\pi^2}{6}$)
4.00	1.082323233711	$\frac{\pi^4}{90}$
3.00	1.202056903160	Apéry's constant
−1.00	−0.083333333333	Ramanujan sum ($-\frac{1}{12}$)
−3.00	0.008333333333	Casimir effect ($\frac{1}{120}$)
0.50	−1.460354508810	Critical line

First ten non-trivial zeros of ζ on critical line are listed in Table 2. Computed t_n match known values to 15 decimal places [10], with $\sigma_n=0.5$. Zero distribution follows RvM formula (21), while the derivative of $N(T)$ gives zero density

$$\frac{dN(T)}{dT}=\frac{1}{2\pi}\log\left(\frac{T}{2\pi}\right)+\mathcal{O}\left(\frac{1}{T}\right), \tag{80}$$

so average spacing follows from (20). Spacing decreases slowly with T and zero density grows logarithmically. RvM formula normalises spacings for MO law and comparison with RMT. Physical constants in Table 3 involve ζ. Planck radiation law gives spectral energy density

Table 2. First 10 non-trivial zeros of ζ on the critical line $\Re(s) = \frac{1}{2}$.

n	t_n	σ_n	n	t_n	σ_n
1	14.134725141734693	0.5	6	37.586178158825668	0.5
2	21.022039638771552	0.5	7	40.918719012147498	0.5
3	25.010857580145689	0.5	8	43.327073280915002	0.5
4	30.424876125859512	0.5	9	48.005150881167161	0.5
5	32.935061587739192	0.5	10	49.773832477672300	0.5

$$u(\omega, T) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\frac{\hbar\omega}{k_B T}} - 1}, \tag{81}$$

with $x = \frac{\hbar\omega}{k_B T}$. Integration over frequencies gives total energy density

$$\begin{aligned} U(T) &= \int_0^\infty u(\omega, T)\, \mathrm{d}\omega = \int_0^\infty \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\hbar\omega/(k_B T)} - 1}\, \mathrm{d}\omega \\ &= \frac{\hbar}{\pi^2 c^3} \left(\frac{k_B T}{\hbar}\right)^4 \int_0^\infty \frac{x^3}{e^x - 1}\, \mathrm{d}x = \frac{\hbar}{\pi^2 c^3} \left(\frac{k_B T}{\hbar}\right)^4 \Gamma(4)\zeta(4), \qquad x = \frac{\hbar\omega}{k_B T}, \end{aligned} \tag{82}$$

where $\Gamma(4) = 3!$, and with standard BE integral

$$\int_0^\infty \frac{x^3}{e^x - 1}\, dx = \frac{\pi^4}{15} = 6\zeta(4), \tag{83}$$

since $\zeta(4) = \frac{\pi^4}{90}$. Thus Boltzmann law

$$\begin{cases} U(T) = \dfrac{\pi^2 k_B^4}{15\hbar^3 c^2} T^4 = \dfrac{4}{c}\sigma T^4, \\ \sigma = \dfrac{\pi^2 k_B^4}{60\hbar^3 c^2} = \dfrac{2\pi^5 k_B^4}{15h^3 c^2}, \end{cases} \tag{84}$$

with σ the Stefan-Boltzmann constant, which recovers the Stefan-Boltzmann law. Thus $\zeta(4)$ emerges in massless boson statistics. Critical temperature T_c for BEC follows from $\zeta\left(\frac{3}{2}\right)$ in (28) while $\zeta\left(\frac{3}{2}\right)$ follows from DOS in 3-d $\int_0^\infty \frac{x^{1/2}}{e^x - 1}\, dx = \frac{\sqrt{\pi}}{2}\zeta\left(\frac{3}{2}\right)$, and $\zeta\left(\frac{3}{2}\right) \approx 2.612$ enters ideal Bose gas thermodynamics. Casimir force per unit area for perfect conductors $F_C = -\frac{\pi^2 \hbar c}{720 a^4}$, with $\frac{\pi^2}{720}$ from $\zeta(-3) = \frac{1}{120}$ via vacuum energy regularisation (13).

Table 3. Physical constants.

Constant	Value	Formula
Boltzmann	5.670374×10^{-8} [$\mathrm{W{\cdot}m^{-2}{\cdot}K^{-4}}$]	$\frac{\pi^2 k_B^4}{60\hbar^3 c^2}$
BEC T_c prefactor	2.612	$\zeta\left(\frac{3}{2}\right)$
Casimir force	1.3×10^{-27} [$\mathrm{N{\cdot}m^{-2}}$]	$\frac{\pi^2 \hbar c}{720 a^4}$

5 Conclusion

This study is consistent with the MO law and HPc via numerical data:

$$\chi^2/\mathrm{dof} = 1.2, \qquad \Delta = \int_0^3 \left| R_2^{\mathrm{num}}(u) - R_2^{\mathrm{GUE}}(u) \right| \mathrm{d}u \approx 0.03, \tag{85}$$

supporting the spectral interpretation of RH. BEC critical temperature $T_c \propto [n/\zeta(3/2)]^{2/3}$ and Casimir energy $E_C \propto \zeta(-3) a^{-3}$ illustrate ζ's role in quantum statistics and QFT [2, 3]. The scattering relation

$$r(t) = \frac{\zeta\left(\frac{1}{2} + it\right) - 1}{\zeta\left(\frac{1}{2} + it\right) + 1}, Z(t) = \frac{1 + r(t)}{1 - r(t)} = \zeta\left(\frac{1}{2} + it\right), Z(t) \sim \frac{1}{\zeta'(\frac{1}{2} + i\gamma_n)(t - \gamma_n)}, \quad t \to \gamma_n, \tag{86}$$

connects zeros to classical wave reflection, with poles at γ_n [16]. Thus RH may be tested in microwave or optical experiments, bridging number theory and laboratory physics.

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A Riemann zeros and gauge symmetries

Spectral, thermodynamic, and scattering properties of Riemann zeros suggest that zeros of ζ may encode GUT symmetry breaking. Zero distribution follows RvM formula (21) and shares RMT statistics of $SO(N)$, $SU(N)$, and $USp(2N)$.

A.1 HP operator and gauge symmetries

HPc proposes that non-trivial zeros of ζ are eigenvalues of a self-adjoint operator. Consider $\mathcal{O}_\alpha$ on weighted Hilbert space $\mathcal{H} = L^2(\mathbb{R}_+, w(T)\,\mathrm{d}T)$ [11]

$$\mathcal{O}_\alpha = -i\left(\frac{\mathrm{d}}{\mathrm{d}T} + \frac{1}{2}\frac{w'(T)}{w(T)}\right) + \alpha V = -i\,w(T)^{-1/2}\frac{\mathrm{d}}{\mathrm{d}T}\,w(T)^{1/2} + \alpha V, \qquad \alpha \in \mathbb{R}, \tag{87}$$

where $w(T) = \frac{\mathrm{d}N(T)}{\mathrm{d}T}$ is zero density from (21)

$$w(T) = \frac{1}{2\pi}\log\left(\frac{T}{2\pi}\right) + \mathcal{O}\left(\frac{1}{T}\right), \tag{88}$$

and V bounded self-adjoint potential encoding zero fluctuations. The spectral theorem gives complete eigenfunctions $\{\psi_n\}$

$$\mathcal{O}_\alpha\psi_n = \lambda_n\psi_n, \quad \langle\psi_m, \psi_n\rangle_{\mathcal{H}} = \delta_{mn}. \tag{89}$$

For specific α_*, V_*, the point spectrum is

$$\sigma_{\mathrm{p}}(\mathcal{O}_{\alpha_*}) = \{\gamma_n : \zeta(1/2 + i\gamma_n) = 0\}. \tag{90}$$

The symmetry of this operator is essential. The logarithmic derivative in (87) links to $SO(N)$ and $SU(N)$ via Cartan decomposition. The Weyl group of $SU(N)$ acts by reflections on roots $\{\alpha_i\}$. Spectral statistics are GUE for $SU(N)$ with $\beta = 2$ and GOE for $SO(N)$ with $\beta = 1$ [19]. The pair correlation $R_2(u)$ in (10) is universal for all β, while higher-order correlations distinguish ensembles [33, 34].

A.2 The Pati-Salam subgroup and pseudo-Riemannian structures

Pati-Salam (PS) model with gauge group $SU(2)_L \times SU(2)_R \times SU(4)$ treats leptons as fourth colour. Fermionic content

$$\psi = \begin{pmatrix} u & d \\ \nu & e \end{pmatrix} \in (2, \bar{2}, 4), \tag{91}$$

with $(2, \bar{2}, 4)$ representation under $SU(2)_L \times SU(2)_R \times SU(4)$. Symmetry breaking to Standard Model (SM)

$$SU(2)_L \times SU(2)_R \times SU(4) \;\longrightarrow\; SU(2)_L \times SU(2)_R \times U(1) \times U(3), \tag{92}$$

via grading operator γ and real structure β. The Dirac operator satisfies

$$D = D_0 + JD_0J^{-1}, \qquad [D, \beta] = 0, \qquad \{D, \gamma\} = 0, \tag{93}$$

with J the real structure, β the grading operator, and γ the chirality operator. Fermionic representation decomposes

$$(2, \bar{2}, 4) \;\longrightarrow\; (2, \bar{2}, 1)_0 \oplus (2, \bar{2}, 3)_1 \oplus (2, \bar{2}, 1)_2, \tag{94}$$

with subscripts $U(1)$ charges. This separates leptons from quarks, preventing leptoquark-mediated proton decay. Symmetry breaking parallels discrete Riemann zero structure. Via the (8) the $\Lambda(n)$ von Mangoldt function, yields peaks at $\log p^k$, mirroring reciprocal relations in gauge symmetry breaking. The PS group embeds into $SO(10)$ via

$$SO(10) \;\supset\; SU(4) \times SU(2)_L \times SU(2)_R, \tag{95}$$

with the 16-dimensional spinor representation decomposing as

$$16 = (4,2,1) \oplus (\bar{4},1,2),\ \dim(16) = 16;\ \text{under } SO(10) \supset SU(4) \times SU(2)_L \times SU(2)_R. \tag{96}$$

The effective potential for Wilson line phases is expressed in terms of ζ-functions

$$V_{\text{eff}}(\theta) = \sum_{n=1}^{\infty} \frac{1}{n} \sum_i \cos(2\pi n \theta_i)\, \zeta(n) = \frac{1}{2} \sum_i \left[\zeta(2) - \zeta(2)\, \theta_i^2 + \mathcal{O}(\theta_i^4)\right], \qquad \theta_i \ll 1. \tag{97}$$

Figure 8 presents spectral statistics of Riemann zeros. Left panel compares spacing distribution with GUE prediction and confirms level repulsion absent in Poisson. Right panel displays first 50 zeros as energy levels of an $SU(N)$ system, with colours indicating representations.

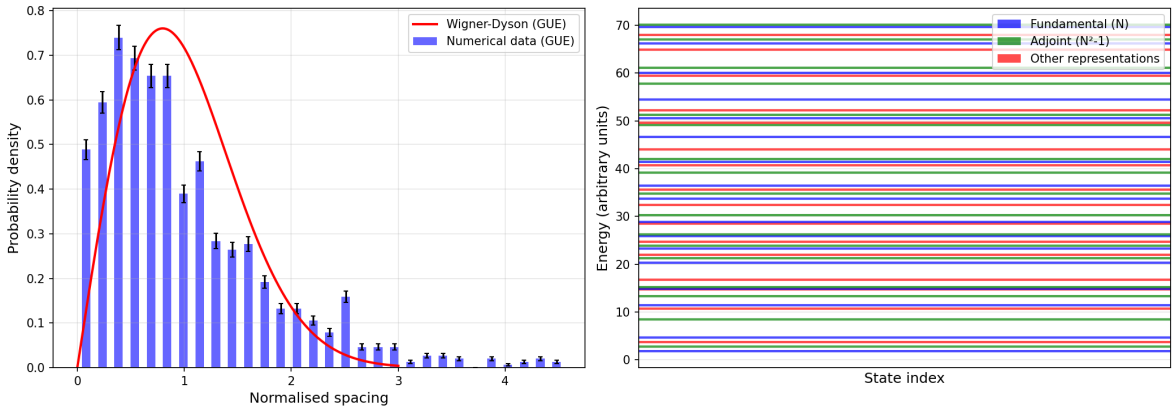


Figure 8. Riemann zeros represented as $SU(N)$ states spacing distribution and the first 50 zeros.

A.3 Schur index and string theory realisation

Schur index of $\mathcal{N} = 4$ supersymmetric Yang-Mills (SYM) with gauge group $SU(3)$ generates divisor sum $\sigma(n)$ [35]

$$\begin{aligned} \mathcal{I}_{\text{S}}(q) &= \sum_{n=1}^{\infty} \sigma(n) q^{2(n-1)} = \prod_{k=1}^{\infty} \frac{1}{1-q^{2k}} \sum_{n=1}^{\infty} \frac{\sigma(n)}{n} q^{2(n-1)} = \frac{1}{(q^2;q^2)_\infty} \sum_{n=1}^{\infty} \frac{\sigma(n)}{n} q^{2(n-1)} \\ &= \text{Tr}_{\mathcal{H}_{\text{BPS}}} (-1)^F q^{2J_3} = \sum_{n=1}^{\infty} d_n q^{2(n-1)} = \sum_{n=1}^{\infty} \frac{q^{2(n-1)}}{1-q^{2n}} = \frac{1}{\eta(\tau)^2} \sum_{n=1}^{\infty} \sigma(n) q^{2(n-1)}, \end{aligned} \tag{98}$$

where $(q;q)_\infty = \prod_{k=1}^{\infty}(1-q^k)$ is Euler function, $\mathcal{H}_{\text{BPS}}$ is the Hilbert space of $\frac{1}{8}$-BPS states, $(-1)^F$ is the fermion number operator, J_3 is the Cartan generator of the $SU(2)$ R-symmetry, $d_n = \sigma(n)$ are the degeneracies, $\eta(\tau) = q^{\frac{1}{24}} \prod_{k=1}^{\infty}(1-q^k)$ is the Dedekind η-function, while index counts $\frac{1}{8}$-BPS bosonic minus fermionic states. Schur index is superconformal index with $t = q$, $p \to 0$

$$\begin{aligned} \mathcal{I}(p,q,t) &= \text{Tr}_{\mathcal{H}} (-1)^F p^{j_1} q^{j_2} t^R \\ &= \sum_{n=1}^{\infty} d_n p^{j_1(n)} q^{j_2(n)} t^{R(n)} = \prod_{k=1}^{\infty} \frac{1}{1-p^{a_k} q^{b_k} t^{c_k}} = \frac{1}{(p;q;t)_\infty} \sum_{n=1}^{\infty} d_n p^{j_1(n)} q^{j_2(n)} t^{R(n)}, \end{aligned} \tag{99}$$

with j_1, j_2 Cartan generators of $SO(4)$ Lorentz group, R R-charge, and $(p;q;t)_\infty$ the generalised Euler function. RH would cancel Kaluza-Klein (KK) modes in supergravity multiplet. Supergravity spectrum on $AdS_5 \times S^5$ has isometry $SO(6) \cong SU(4)$; KK masses

$$m^2 = \frac{1}{R^2} \left(\sum_{i=1}^{3} n_i (n_i + 4 - 2i) + \cdots \right), \tag{100}$$

with R radius of S^5 and n_i mode numbers. Supergravity spectrum

$$\begin{aligned} \text{Graviton:}\quad m^2 = 0, \qquad \text{Dilaton:}\quad m^2 = \frac{4}{R^2}, \\ \text{Vector:}\quad m^2 = \frac{4}{R^2}, \qquad \text{Scalar:}\quad m^2 = \frac{\Delta(\Delta-4)}{R^2}, \end{aligned} \tag{101}$$

with $\Delta = 2, 3, 4, \ldots$ KK levels. $\mathrm{AdS}_5/\mathrm{CFT}_4$ duality maps $SU(N)$ to S^5 geometry. Zeros of ζ are spectrum of dilation operator in boundary CFT, mirroring dual string theory levels. In AdS_3/CFT_2, modular invariance relates primary spectrum to zeros of ζ

$$Z_{\mathrm{CFT}}(\tau) = \sum_n d_n q^n = \prod_{n=1}^{\infty}(1-q^n)^{-1} \sum_{n=1}^{\infty} \frac{\sigma(n)}{n} q^n = q^{-\frac{c}{24}} \prod_{n=1}^{\infty}(1-q^n)^{-1}, \qquad q = e^{2\pi i \tau}, \tag{102}$$

with d_n CFT degeneracies and $q = e^{2\pi i\tau}$. Modular transformation $\tau \to -\frac{1}{\tau}$ gives the functional equation

$$Z_{\mathrm{CFT}}\left(-\frac{1}{\tau}\right) = \tau^{\frac{c}{2}} Z_{\mathrm{CFT}}(\tau) = \tau^{\frac{c}{2}} \sum_n d_n q^n = \tau^{\frac{c}{2}} \prod_{n=1}^{\infty}(1-q^n)^{-1} \sum_{n=1}^{\infty} \frac{\sigma(n)}{n} q^n, \tag{103}$$

with c central charge. Poles of $Z_{\mathrm{CFT}}(\tau)$ in complex τ-plane follow from zeros of ζ as

$$Z_{\mathrm{CFT}}(\tau) = \sum_n d_n q^n = \prod_{n=1}^{\infty}(1-q^n)^{-1} \sum_{n=1}^{\infty} \frac{\sigma(n)}{n} q^n = \sum_{n=1}^{\infty} \frac{q^n}{1-q^n} = \frac{1}{\eta(\tau)} \sum_{n=1}^{\infty} \sigma(n) q^n, \tag{104}$$

where d_n are the CFT degeneracies, $q = e^{2\pi i\tau}$, and $\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)$ is the Dedekind eta function. As $\tau \to \frac{i}{2}$, partition function has the asymptotic

$$Z_{\mathrm{CFT}}(\tau) \sim \exp\left(\frac{\pi i c}{12\tau}\right) + \cdots, \tag{105}$$

typical of large-c CFT. Poles are zeros of ζ via spectral RH

$$\zeta\left(\frac{1}{2} + i\gamma_n\right) = 0 \quad \Longleftrightarrow \quad \tau_n = \frac{i}{2} + \frac{1}{\gamma_n} + \cdots. \tag{106}$$

RH is equivalent to poles lying on $\Re(\tau) = \frac{1}{2}$, the modular image of the critical line.

A.4 Symmetry breaking and the zeros

Symmetry breaking of $SU(N)$

$$SU(N) \to SU(N_1) \times SU(N_2) \times \cdots \times U(1)^r, \tag{107}$$

with Wilson phases θ_i. Effective potential (97) expands as $V_{\mathrm{eff}}(\theta) = \frac{1}{2}\sum_i[\zeta(2) - \zeta(2)\theta_i^2 + \cdots]$, giving gauge boson masses $m^2 \propto \zeta(2)$. Minima

$$\frac{\partial V_{\mathrm{eff}}}{\partial \theta_i} = 0, \qquad \frac{\partial^2 V_{\mathrm{eff}}}{\partial \theta_i^2} > 0, \tag{108}$$

determine breaking, while ζ regularises KK sums

$$\sum_{n=1}^{\infty} \frac{1}{n} \cos(2\pi n\theta_i)\zeta(n) = -\log\big(2\sin(\pi\theta_i)\big) + \cdots, \tag{109}$$

with critical temperatures

$$\begin{aligned} T_c \propto \left(\frac{1}{\zeta(n)}\right)^{\frac{1}{n-1}} = \zeta(n)^{-\frac{1}{n-1}} = \exp\left[-\frac{\log\zeta(n)}{n-1}\right] = \left(\sum_{k=1}^{\infty}\frac{1}{k^n}\right)^{-\frac{1}{n-1}} \\ = \left(\prod_p \frac{1}{1-p^{-n}}\right)^{-\frac{1}{n-1}} = \prod_p \left(1-p^{-n}\right)^{\frac{1}{n-1}}, \quad \text{with} \;\; n \geq 2, \end{aligned} \tag{110}$$

The PS group breaks through the chain

$$SO(10) \xrightarrow{M_{\rm GUT}} SU(4) \times SU(2)_L \times SU(2)_R \xrightarrow{M_{\rm PS}} SU(3)_c \times SU(2)_L \times U(1)_Y \xrightarrow{M_{\rm EW}} SU(3)_c \times U(1)_{\rm EM}, \tag{111}$$

with $M_{\rm GUT} \sim 10^{16}$ GeV, $M_{\rm PS} \sim 10^{12}$ GeV, and $M_{\rm EW} \sim 10^2$ GeV the intermediate scales, and $SU(3)_c \times U(1)_{\rm EM}$ the unbroken gauge group at low energies. while the RG flow

$$\frac{dg_i}{d\ln\mu} = \frac{b_i}{16\pi^2} g_i^3 + \sum_{j,k} \frac{b_{ijk}}{(16\pi^2)^2} g_i g_j^2 g_k^2 + \cdots, \tag{112}$$

has fixed points

$$\begin{aligned} \beta(g_*) = 0 \iff \left.\frac{dg_i}{d\ln\mu}\right|_{g_i=g_{i*}} = 0 \iff \zeta\left(\frac{1}{2} + i\gamma_n\right) = 0 \\ \iff \gamma_n \in \mathbb{R} \iff \mathcal{O}_\alpha = \mathcal{O}_\alpha^\dagger, \qquad n \in \mathbb{N}, \end{aligned} \tag{113}$$

The scale hierarchy is encoded in the zeros of ζ through the RG flow (112), where the beta function coefficients b_i receive KK contributions. RH corresponds to no unwanted fixed points (113). Values of $\zeta(n)$ listed in Table 4 appear in effective potentials of gauge theories, as shown in (97). Their physical relevance extends from black-body radiation to $SU(4)$ breaking in the PS model. The critical temperature for PS breaking is

$$T_{\rm PS} \sim \frac{1}{R}\sqrt{\frac{1}{\zeta(2)}} \sim \frac{1}{R}, \tag{114}$$

which links ζ directly to the symmetry breaking scale.

Table 4. Values of $\zeta(n)$ and their physical relevance.

n	$\zeta(n)$	Context	Domain
2	1.644934066848	KK sums	black body radiation
3	1.202056903160	BEC critical temperature	AdS/CFT
4	1.082323233711	BC	QED, black body radiation
-1	-0.083333333333	Zeta regularisation	Casimir effect
-3	0.008333333333	Casimir force	PS breaking
0.5	-1.460354508810	Critical line	spectral statistics

A.5 Zeros as a physical spectrum

Riemann zeros admit multiple consistent physical interpretations across different areas of theoretical physics, unified by the spectral relation (113), where $\mathcal{O}_\alpha$ is the self-adjoint operator (87) and γ_n are the imaginary parts of the non-trivial zeros. The corresponding spectral statistics follow the GUE form (10), with number variance (55) and spectral rigidity (56). These relations connect the zeros to quantum chaos, gauge symmetries, and string theory, as detailed below.

A.5.1 Spectral realisation via the HP operator The spectral interpretation of zeros (see Introduction) is realised via the HP operator. A prime-based Hilbert space construction uses primes as eigenstates and entropy-driven stabilisation for spectral alignment [36]

$$\mathcal{H}_P = \ell^2(\mathbb{P}, w) \oplus \ell^2(\mathbb{P}, w) = \left\{ (a_p)_{p\in\mathbb{P}} : \sum_p w(p)|a_p|^2 < \infty \right\}, \tag{115}$$

with weight function

$$w(p) = \frac{\log p}{p^{1+\alpha}} = \frac{\log p}{p} \cdot p^{-\alpha}, \qquad \alpha > 0, \tag{116}$$

ensuring square summability. Block chiral operator

$$
\begin{aligned}
H = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix}, \qquad \Gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \{\Gamma, H\} = 0, \\
H^2 = \begin{pmatrix} A^\dagger A & 0 \\ 0 & AA^\dagger \end{pmatrix}, \qquad \Gamma^2 = I, \qquad \operatorname{Tr} H = 0, \\
\operatorname{spec}(H) = \operatorname{spec}(A^\dagger A)^{1/2} \cup \operatorname{spec}(AA^\dagger)^{1/2},
\end{aligned}
\tag{117}
$$

with kernel

$$
\begin{aligned}
A_{pq} &= r_{pq} + \sum_{m\geq 1} \frac{\log p}{p^{\frac{m}{2}}} u_m(p) \cos(m \log p\, \phi_q) = r_{pq} + \sum_{m\geq 1} \frac{\log p}{p^{\frac{m}{2}}} u_m(p) \left[e^{im \log p\, \phi_q} + e^{-im \log p\, \phi_q} \right] \\
&= r_{pq} + 2 \sum_{m\geq 1} \frac{\log p}{p^{\frac{m}{2}}} u_m(p) \cos(m \log p\, \phi_q), \quad r_{pq} = r_{qp},
\end{aligned}
\tag{118}
$$

where $r_{pq} = r_{qp}$ decays as $\mathcal{O}\big((pq)^{-1-\epsilon}\big)$ and phases $\phi_q = c_1 \log q$ ensure spectral coherence. With these assumptions, A is Hilbert-Schmidt and H self-adjoint with compact resolvent [36]. Spectral determinant of H

$$
\Delta_H(t) = \prod_{n\geq 1} \frac{\lambda_n^2 + t^2}{\lambda_n^2 + 1} = \exp\left[\sum_{n\geq 1} \log\left(\frac{\lambda_n^2 + t^2}{\lambda_n^2 + 1} \right) \right], \qquad t \in \mathbb{R}, \tag{119}
$$

equals completed Riemann ξ-function on critical line

$$
\frac{d}{dt} \log \Delta_H(t) = \frac{d}{dt} \log \xi\left(\frac{1}{2} + it \right). \tag{120}
$$

Non-trivial zeros appear as spectral singularities of resolvent [36]. Alternatively, similarity transformation of Berry-Keating Hamiltonian

$$
\begin{aligned}
\hat{H} = e^{x/2} \hat{\tilde{H}} e^{-x/2} &= -\hat{D} - i \sum_{m=0}^{\infty} c_m \hat{T}^m = -\frac{1}{2} (x\hat{p} + \hat{p}x) - i \sum_{m=0}^{\infty} c_m \left(x\hat{p}^2 + \hat{p}^2 x \right)^m \\
&= -i \left(x \frac{\mathrm{d}}{\mathrm{d}x} + \frac{1}{2} \right) + \sum_{m=1}^{\infty} c_m \hat{T}^m, \qquad \hat{T} = x\hat{p}^2 + \hat{p}^2 x,
\end{aligned}
\tag{121}
$$

with $\hat{D} = \frac{x\hat{p} + \hat{p}x}{2}$ and $\hat{T} = \frac{x\hat{p}^2 + \hat{p}^2 x}{2}$ Bessel operator [11]. Eigenvalue equation for $\hat{T}$ is Bessel's equation

$$
\left(-x \frac{d^2}{dx^2} - \frac{d}{dx} \right) \langle x|t\rangle = t \langle x|t\rangle, \tag{122}
$$

with solutions $\langle x|t\rangle = J_0(2\sqrt{xt})$, $t \in \mathbb{R}^+$. Eigenfunctions $\widetilde{\Psi}_s(x)$ vanish at origin under boundary condition

$$
\begin{aligned}
\widetilde{\Psi}_s(0) &= \left(1 - 2^{1-s}\right) \Gamma(s)\, \zeta(s) = \left(1 - 2^{1-s}\right) \Gamma(s) \sum_{n=1}^{\infty} \frac{1}{n^s} \\
&= \left(1 - 2^{1-s}\right) \Gamma(s) \prod_p \left(1 - p^{-s}\right)^{-1} = 0 \\
&\iff \quad \zeta(s) = 0 \quad \iff \quad s = \frac{1}{2} + i\gamma_n, \qquad n \in \mathbb{N},
\end{aligned}
\tag{123}
$$

linking spectral problem to zeros of ζ [11].

A.5.2 Diffraction grating and prime optics Zeros γ_n realised physically via aperiodic diffraction grating, where slit positions are set to γ_n. The slit function is

$$g(x) = \sum_{n=1}^{N} \delta(x - \gamma_n) = \sum_{n=1}^{N} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(x-\gamma_n)} \, \mathrm{d}\omega, \tag{124}$$

with the grating transmission function

$$T(x) = \sum_{n=1}^{N} \mathrm{rect}\left(\frac{x - \gamma_n}{w}\right), \qquad \mathrm{rect}(u) = \begin{cases} 1, & |u| \le \frac{1}{2}, \\ 0, & |u| > \frac{1}{2}, \end{cases} \tag{125}$$

where w is the slit width. The diffraction amplitude in the far field is

$$\begin{aligned} S_N(t) &= 2\sum_{n=1}^{N} \cos(\gamma_n t) = \sum_{n=1}^{N} \left(e^{i\gamma_n t} + e^{-i\gamma_n t}\right) = 2\,\Re \sum_{n=1}^{N} e^{i\gamma_n t} = 2\,\Re\,\mathrm{Tr}\left(e^{it\mathcal{O}_\alpha}\right) \\ &= \mathrm{Tr}\left(e^{it\mathcal{O}_\alpha} + e^{-it\mathcal{O}_\alpha}\right) = 2\sum_{n=1}^{N} \cos(\gamma_n t) = 2\,\Re \sum_{n=1}^{N} \left(e^{it}\right)^{\gamma_n}, \qquad \mathcal{O}_\alpha \psi_n = \gamma_n \psi_n, \end{aligned} \tag{126}$$

where $\mathcal{O}_\alpha$ is the self-adjoint operator (87) with eigenvalues γ_n. The Fourier transform is

$$\begin{aligned} \tilde{S}_N(\omega) &= \int_{-\infty}^{\infty} S_N(t) e^{-i\omega t} \, \mathrm{d}t = \sum_{n=1}^{N} \left[\delta(\omega - \gamma_n) + \delta(\omega + \gamma_n)\right] \\ &= \sum_{n=1}^{N} \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(e^{i\gamma_n t} + e^{-i\gamma_n t}\right) e^{-i\omega t} \, \mathrm{d}t = \frac{1}{2\pi} \sum_{n=1}^{N} \left[\delta(\omega - \gamma_n) + \delta(\omega + \gamma_n)\right], \end{aligned} \tag{127}$$

with the inverse transform

$$\begin{aligned} S_N(t) &= \int_{-\infty}^{\infty} \tilde{S}_N(\omega) e^{i\omega t} \, \frac{\mathrm{d}\omega}{2\pi} = \frac{1}{2\pi} \sum_{n=1}^{N} \int_{-\infty}^{\infty} \left[\delta(\omega - \gamma_n) + \delta(\omega + \gamma_n)\right] e^{i\omega t} \, \mathrm{d}\omega \\ &= \frac{1}{2\pi} \sum_{n=1}^{N} \left(e^{i\gamma_n t} + e^{-i\gamma_n t}\right) = \frac{1}{\pi} \sum_{n=1}^{N} \cos(\gamma_n t) = \frac{1}{2\pi} \mathrm{Tr}\left(e^{it\mathcal{O}_\alpha} + e^{-it\mathcal{O}_\alpha}\right) \\ &= \frac{1}{\pi} \mathrm{Tr}\left[\cos(t\mathcal{O}_\alpha)\right]. \qquad t \in \mathbb{R}, \end{aligned} \tag{128}$$

The power spectrum is

$$\begin{aligned} |\tilde{S}_N(\omega)|^2 = \tilde{S}_N(\omega)\tilde{S}_N^*(\omega) &= \sum_{n,m=1}^{N} \left[\delta(\omega - \gamma_n)\delta(\omega - \gamma_m) + \delta(\omega + \gamma_n)\delta(\omega + \gamma_m)\right] \\ &= \sum_{n=1}^{N} \left[\delta^2(\omega - \gamma_n) + \delta^2(\omega + \gamma_n)\right] + 2\sum_{n \ne m} \delta(\omega - \gamma_n)\delta(\omega - \gamma_m) \\ &= \sum_{n=1}^{N} \left[\delta(\omega - \gamma_n) + \delta(\omega + \gamma_n)\right] = \rho_N(\omega) + \rho_N(-\omega), \end{aligned} \tag{129}$$

where $\rho_N(\omega) = \sum_{n=1}^{N} \delta(\omega - \gamma_n)$ is the spectral density, while the corresponding autocorrelation function is

$$\begin{aligned} C_N(\tau) &= \int_{-\infty}^{\infty} S_N(t) S_N(t + \tau) \, \mathrm{d}t = \sum_{n=1}^{N} \cos(\gamma_n \tau) = \sum_{n=1}^{N} \frac{1}{2} \left(e^{i\gamma_n \tau} + e^{-i\gamma_n \tau}\right) \\ &= \frac{1}{2} \mathrm{Tr}\left(e^{i\tau\mathcal{O}_\alpha} + e^{-i\tau\mathcal{O}_\alpha}\right) = \mathrm{Tr}\left[\cos(\tau\mathcal{O}_\alpha)\right] = \sum_{n=1}^{N} \cos(\gamma_n \tau), \qquad \tau \in \mathbb{R}, \end{aligned} \tag{130}$$

The cumulative spectral function is

$$
\begin{aligned}
F_N(\omega) &= \int_{-\infty}^{\omega} |\tilde{S}_N(\omega')|^2 \,\mathrm{d}\omega' = \int_{-\infty}^{\omega} \sum_{n=1}^{N} \delta(\omega' - \gamma_n)\,\mathrm{d}\omega' = \sum_{n=1}^{N} \int_{-\infty}^{\omega} \delta(\omega' - \gamma_n)\,\mathrm{d}\omega' \\
&= \sum_{n=1}^{N} \Theta(\omega - \gamma_n) = \sum_{n=1}^{N} \frac{1}{2}\left[1 + \operatorname{sgn}(\omega - \gamma_n)\right] = \frac{N}{2} + \frac{1}{2}\sum_{n=1}^{N} \operatorname{sgn}(\omega - \gamma_n) \\
&= \frac{N}{2} + \frac{1}{\pi}\sum_{n=1}^{N} \arctan\left(\frac{\omega - \gamma_n}{\varepsilon}\right) + \mathcal{O}(\varepsilon), \qquad \varepsilon \to 0^+,
\end{aligned} \tag{131}
$$

where Θ is the Heaviside step function. The spectral density is

$$
\begin{aligned}
\rho_N(\omega) &= \frac{\mathrm{d}F_N(\omega)}{\mathrm{d}\omega} = \sum_{n=1}^{N} \delta(\omega - \gamma_n) = \sum_{n=1}^{N} \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(\omega-\gamma_n)t}\,\mathrm{d}t \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\omega t} \sum_{n=1}^{N} e^{-i\gamma_n t}\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} \operatorname{Tr}\left(e^{it(\omega I - \mathcal{O}_\alpha)}\right)\,\mathrm{d}t \\
&= \frac{1}{2\pi}\operatorname{Tr}\int_{-\infty}^{\infty} e^{it(\omega I - \mathcal{O}_\alpha)}\,\mathrm{d}t = \operatorname{Tr}\delta(\omega I - \mathcal{O}_\alpha), \qquad \omega \in \mathbb{R},
\end{aligned} \tag{132}
$$

The moments of the spectral density are

$$
M_k = \int_{-\infty}^{\infty} \omega^k \rho_N(\omega)\,\mathrm{d}\omega = \sum_{n=1}^{N} \gamma_n^k, \qquad k \in \mathbb{N}. \tag{133}
$$

The spectral form factor is

$$
\begin{aligned}
K_N(\tau) &= \frac{1}{N}\left|\sum_{n=1}^{N} e^{i\gamma_n\tau}\right|^2 = \frac{1}{N}\sum_{n,m=1}^{N} e^{i(\gamma_n-\gamma_m)\tau} = \frac{1}{N}\sum_{n=1}^{N} e^{i(\gamma_n-\gamma_n)\tau} + \frac{1}{N}\sum_{n\neq m} e^{i(\gamma_n-\gamma_m)\tau} \\
&= 1 + \frac{1}{N}\sum_{n\neq m} e^{i(\gamma_n-\gamma_m)\tau} = 1 + \frac{2}{N}\sum_{n<m} \cos[(\gamma_n - \gamma_m)\tau] \\
&= 1 + \frac{2}{N}\sum_{k=1}^{N-1}\sum_{n=1}^{N-k} \cos[(\gamma_{n+k} - \gamma_n)\tau]\,, \qquad \tau \in \mathbb{R},
\end{aligned} \tag{134}
$$

with the ensemble average

$$
\begin{aligned}
\langle K_N(\tau)\rangle &= \left\langle \frac{1}{N}\left|\sum_{n=1}^{N} e^{i\gamma_n\tau}\right|^2 \right\rangle = 1 + \frac{1}{N}\sum_{n\neq m} \left\langle e^{i(\gamma_n-\gamma_m)\tau}\right\rangle \\
&= 1 + \frac{1}{N}\sum_{n\neq m}\int_{-\infty}^{\infty} e^{i(\gamma_n-\gamma_m)\tau}\,P(\gamma_n,\gamma_m)\,\mathrm{d}\gamma_n\,\mathrm{d}\gamma_m \\
&= 1 + \frac{1}{N}\sum_{n\neq m}\left[\int_{-\infty}^{\infty} e^{i\gamma_n\tau}P(\gamma_n)\,\mathrm{d}\gamma_n\right]\left[\int_{-\infty}^{\infty} e^{-i\gamma_m\tau}P(\gamma_m)\,\mathrm{d}\gamma_m\right] \\
&= 1 + \frac{1}{N}\sum_{n\neq m}\left|\tilde{P}(\tau)\right|^2 = 1 + \frac{N-1}{N}\left|\tilde{P}(\tau)\right|^2, \qquad \tau \in \mathbb{R},
\end{aligned} \tag{135}
$$

where $P(\gamma_n)$ is the probability density of the zeros and $\tilde{P}(\tau) = \int_{-\infty}^{\infty} e^{i\gamma\tau}P(\gamma)\,\mathrm{d}\gamma$ is its Fourier transform. For large N (8) relates $S_N(t)$ to the von Mangoldt function $\Lambda(n)$ [37]

$$
\begin{aligned}
S_N(t) &\approx \frac{2\pi N}{\gamma_N}\frac{\sin(\gamma_N t)}{\pi t} - \frac{\mathrm{Si}(\gamma_N t)}{\pi t} - \sum_p \frac{\Lambda(p)}{\sqrt{p}}\frac{\sin[\gamma_N(t\pm\ln p)]}{\pi(t\pm\ln p)} \\
&= \frac{2N}{\gamma_N t}\sin(\gamma_N t) - \frac{\mathrm{Si}(\gamma_N t)}{\pi t} - \sum_p \frac{\log p}{\sqrt{p}}\frac{\sin[\gamma_N(t\pm\ln p)]}{\pi(t\pm\ln p)} \\
&= \frac{2N}{\gamma_N t}\sin(\gamma_N t) - \frac{1}{\pi t}\int_0^{\gamma_N t}\frac{\sin u}{u}\,\mathrm{d}u - \sum_p \frac{\log p}{\sqrt{p}}\frac{\sin[\gamma_N(t\pm\ln p)]}{\pi(t\pm\ln p)} \\
&= \frac{2N}{\gamma_N t}\sin(\gamma_N t) - \frac{1}{2\pi i t}\left[E_1(-i\gamma_N t) - E_1(i\gamma_N t)\right] - \sum_p \frac{\log p}{\sqrt{p}}\frac{\sin[\gamma_N(t\pm\ln p)]}{\pi(t\pm\ln p)} \\
&= \frac{2N}{\gamma_N t}\sin(\gamma_N t) - \frac{1}{\pi t}\,\mathrm{Im}\left[E_1(i\gamma_N t)\right] - \sum_p \frac{\log p}{\sqrt{p}}\frac{\sin[\gamma_N(t\pm\ln p)]}{\pi(t\pm\ln p)} \\
&= \frac{2N}{\gamma_N t}\sin(\gamma_N t) - \frac{1}{\pi t}\,\mathrm{Im}\left[\Gamma(0,-i\gamma_N t)\right] - \sum_p \frac{\log p}{\sqrt{p}}\frac{\sin[\gamma_N(t\pm\ln p)]}{\pi(t\pm\ln p)} \\
&= \frac{2N}{\gamma_N t}\sin(\gamma_N t) - \frac{1}{\pi t}\,\mathrm{Im}\left[\int_{i\gamma_N t}^{\infty}\frac{e^{-u}}{u}\,\mathrm{d}u\right] - \sum_p \frac{\log p}{\sqrt{p}}\frac{\sin[\gamma_N(t\pm\ln p)]}{\pi(t\pm\ln p)}, \qquad t\to\infty,
\end{aligned}
\tag{136}
$$

where $\mathrm{Si}(x)=\int_0^x \frac{\sin u}{u}\,\mathrm{d}u$ is the sine integral, $E_1(z)=\int_z^\infty \frac{e^{-u}}{u}\,\mathrm{d}u$ is the exponential integral, $\Gamma(0,z)=\int_z^\infty \frac{e^{-u}}{u}\,\mathrm{d}u$ is the incomplete gamma function, and $\Lambda(p)=\log p$ is the von Mangoldt function for primes. Taking into account (65), the resolution improves with N. The signal-to-noise ratio is

$$
\begin{aligned}
\mathrm{SNR} &= \frac{|S_N(t_{\mathrm{peak}})|^2}{\sigma^2} \propto N^2\left(\frac{\log p}{p}\right)^2 = N^2\frac{I(p)}{p} \\
&= N^2\frac{1}{p}\left|\tilde{S}_N(\omega_p)\right|^2 = N^2\frac{1}{p}\left|\sum_{n=1}^{N} e^{i\gamma_n\omega_p}\right|^2, \qquad \omega_p=\log p,
\end{aligned}
\tag{137}
$$

where σ is the noise level. The diffraction efficiency is

$$
\begin{aligned}
\eta &= \frac{\int_{\mathrm{peak}}|\tilde{S}_N(\omega)|^2\,\mathrm{d}\omega}{\int_{-\infty}^{\infty}|\tilde{S}_N(\omega)|^2\,\mathrm{d}\omega} = \frac{\sum_{n=1}^N\int_{\mathrm{peak}}\delta(\omega-\gamma_n)\,\mathrm{d}\omega}{\sum_{n=1}^N\int_{-\infty}^{\infty}\delta(\omega-\gamma_n)\,\mathrm{d}\omega} = \frac{\sum_{n=1}^N\Theta(\omega_{\mathrm{peak}}-\gamma_n)}{N} \\
&= \frac{1}{N}\,\mathrm{Tr}\left[\Theta(\omega_{\mathrm{peak}}I-\mathcal{O}_\alpha)\right] = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{2}\left[1+\mathrm{sgn}(\omega_{\mathrm{peak}}-\gamma_n)\right], \qquad \omega_{\mathrm{peak}}\in\mathbb{R}.
\end{aligned}
\tag{138}
$$

The resolving power, grating equation, and diffraction order are collected as,

$$
\begin{cases}
R = \dfrac{\lambda}{\Delta\lambda} = \gamma_N\Delta t \sim 1, \\
d\sin\theta = m\lambda, \qquad m\in\mathbb{Z}, \\
m = \dfrac{d\sin\theta}{\lambda} = \dfrac{\gamma_N\Delta t}{2\pi} \sim \dfrac{1}{2\pi},
\end{cases}
\tag{139}
$$

respectively, where R is the resolving power, λ the wavelength, $\Delta\lambda$ the wavelength resolution, γ_N the N-th zero, Δt the peak width, d the grating period, θ the diffraction angle, and m the diffraction order.This provides a direct experimental realisation of the Riemann zeros in laboratory settings, complementing the scattering platform in Sec. 3.7 where zeros appear as resonances in $r(t)$ and $Z(t)$ (77). Together, these offer a physical pathway to test the RH via microwave cavities or optical systems [37].

A.5.3 AdS/CFT correspondence and string theory In AdS/CFT, zeros of ζ emerge as spectrum of dilation operator in boundary CFT [35]. The Schur index of $\mathcal{N}=4$ SYM with gauge group $SU(N)$ generates the divisor sum $\sigma(n)$

$$\begin{aligned}\mathcal{I}_S(q) &= \sum_{n=1}^{\infty}\sigma(n)q^{2(n-1)} = \sum_{n=1}^{\infty}\frac{q^{2(n-1)}}{1-q^{2n}} = \prod_{k=1}^{\infty}\frac{1}{1-q^{2k}}\sum_{n=1}^{\infty}\frac{\sigma(n)}{n}q^{2(n-1)}\\ &= \frac{1}{(q^2;q^2)_\infty}\sum_{n=1}^{\infty}\frac{\sigma(n)}{n}q^{2(n-1)} = \mathrm{Tr}_{\mathcal{H}_{\rm BPS}}(-1)^F q^{2J_3} = \sum_{n=1}^{\infty}d_n q^{2(n-1)}\\ &= \frac{1}{\eta(\tau)^2}\sum_{n=1}^{\infty}\sigma(n)q^{2(n-1)} = \sum_{n=1}^{\infty}\frac{q^{2(n-1)}}{1-q^{2n}} = \sum_{n=1}^{\infty}\sum_{k=1}^{\infty}q^{2k(n-1)} = \sum_{k=1}^{\infty}\frac{q^{2(k-1)}}{1-q^{2k}}\\ &= \sum_{k=1}^{\infty}\frac{1}{q^{-2(k-1)}-q^2} = \frac{1}{(q^2;q^2)_\infty}\sum_{n=1}^{\infty}\frac{\sigma(n)}{n}q^{2(n-1)} = \frac{1}{\eta(\tau)^2}\sum_{n=1}^{\infty}\sigma(n)q^{2(n-1)},\end{aligned} \tag{140}$$

where $q = e^{2\pi i\tau}$ with the large-N expansion

$$\mathcal{I}_S^{(N)}(q) = \frac{1}{(q;q)_\infty}\sum_{n=1}^{\infty}\frac{\sigma(n)}{N}q^{2(n-1)} + \mathcal{O}(q^{2N}), \tag{141}$$

and $(q;q)_\infty = \prod_{k=1}^{\infty}(1-q^k)$ is the Euler function, while the $\mathrm{AdS}_5/\mathrm{CFT}_4$ duality maps this gauge theory to type IIB superstring theory on $AdS_5 \times S^5$, with the boundary CFT PF in (104) having $q = e^{2\pi i\tau}$ and d_n degeneracies of CFT primaries, so that modular invariance of the CFT implies the functional equation

$$\begin{aligned}Z_{\rm CFT}\left(-\frac{1}{\tau}\right) &= \tau^{c/2}Z_{\rm CFT}(\tau) = \tau^{c/2}\sum_n d_n q^n = \tau^{c/2}\prod_{n=1}^{\infty}(1-q^n)^{-1}\sum_{n=1}^{\infty}\frac{\sigma(n)}{n}q^n\\ &= \tau^{c/2}\frac{1}{\eta(\tau)}\sum_{n=1}^{\infty}\sigma(n)q^n, \qquad q = e^{2\pi i\tau},\end{aligned} \tag{142}$$

where c is the central charge. Poles of $Z_{\rm CFT}(\tau)$ in the complex τ-plane correspond to zeros of ζ through the relation

$$\begin{aligned}\zeta\left(\frac{1}{2}+i\gamma_n\right) = 0 \quad &\Longleftrightarrow \quad \tau_n = \frac{i}{2}+\frac{1}{\gamma_n}+\cdots \quad \Longleftrightarrow \quad Z_{\rm CFT}(\tau_n)\to\infty\\ &\Longleftrightarrow \quad \mathrm{Re}(\tau_n) = \frac{1}{2}, \qquad n\in\mathbb{N},\end{aligned} \tag{143}$$

This is the holographic dual of the spectral interpretation of RH, which is equivalent to all poles of the CFT PF lying on the line $\Re(\tau) = \frac{1}{2}$ [35]. Corresponding cancellation of KK modes in (100) under RH would imply a relation between the zeros and the spectrum of the boundary dilation operator, mirroring the statistical properties of chaotic quantum systems.

A.5.4 Unified perspective: gauge symmetries, RG flows, and RH The formula (8) above encodes the duality between zeros and primes, connecting zero statistics GUE (11) to the prime distribution, while the RH is equivalent to $\gamma_n \in \mathbb{R}$, i.e. self-adjointness of $\mathcal{O}_\alpha$ (87). The PS model provides a gauge-theoretic realisation with the breaking chain

$$\begin{aligned}SO(10) &\xrightarrow{M_{\rm GUT}} SU(4)\times SU(2)_L\times SU(2)_R \xrightarrow{M_{\rm PS}} SU(3)_c\times SU(2)_L\times U(1)_Y\\ &\xrightarrow{M_{\rm EW}} SU(3)_c\times U(1)_{\rm EM} \xrightarrow{M_{\rm QCD}} SU(3)_c\\ &= SU(4)\times SU(2)_L\times SU(2)_R \xrightarrow{\langle\Phi\rangle} SU(3)_c\times SU(2)_L\times U(1)_Y\\ &= SU(5) \xrightarrow{\langle\Phi\rangle} SU(3)_c\times SU(2)_L\times U(1)_Y\\ &= SO(10) \xrightarrow{\langle\Phi\rangle} SU(5), \qquad G_{\rm PS} = SU(4)\times SU(2)_L\times SU(2)_R,\end{aligned} \tag{144}$$

where $\langle\Phi\rangle$ denotes the vacuum expectation value of the scalar field Φ, and $M_{\rm GUT}\sim 10^{16}$ GeV, $M_{\rm PS}\sim 10^{12}$ GeV, $M_{\rm EW}\sim 10^2$ GeV, and $M_{\rm QCD}\sim 10^{-1}$ GeV are the symmetry breaking scales, and $G_{\rm PS} = SU(2)_L\times SU(2)_R\times U(1)\times U(3)$. The effective potential (97) has minima at

$$\begin{aligned}\frac{\partial V_{\text{eff}}}{\partial\theta_i}=0,\qquad \frac{\partial^2 V_{\text{eff}}}{\partial\theta_i^2}>0&=\sum_{n=1}^{\infty}\frac{1}{n}\sum_j\sin(2\pi n\theta_j)\zeta(n)=0\\ &=\sum_{n=1}^{\infty}\sum_j\sin(2\pi n\theta_j)\zeta(n)=0,\qquad \theta_i\in[0,1),\end{aligned}\tag{145}$$

determining the vacuum structure. The RG flow (112) has fixed points

$$\begin{aligned}\beta(g_*)=0\quad&\Longleftrightarrow\quad \zeta\Big(\frac{1}{2}+i\gamma_n\Big)=0\\ \Longleftrightarrow\quad&\frac{dg_i}{d\ln\mu}\Big|_{g_i=g_{i*}}=0\quad\Longleftrightarrow\quad\gamma_n\in\mathbb{R}\quad\Longleftrightarrow\quad\mathcal{O}_\alpha=\mathcal{O}_\alpha^\dagger,\qquad n\in\mathbb{N},\end{aligned}\tag{146}$$

so RH corresponds to the absence of unwanted fixed points [19, 38], while the pseudo-Riemannian structure of the PS model breaks $SU(4)$ to $U(1)\times U(3)$, mirroring how zeros break conformal symmetry on the critical line and yielding the discrete set $\{\gamma_n\}$. This duality is encoded in the spectral determinant (119), which matches the Riemann ξ-function

$$\begin{aligned}\frac{d}{dt}\log\Delta_H(t)=\frac{d}{dt}\log\xi\Big(\frac{1}{2}+it\Big)&=\frac{\xi'\big(\frac{1}{2}+it\big)}{\xi\big(\frac{1}{2}+it\big)}=\frac{d}{dt}\log\prod_{n=1}^{\infty}\frac{\lambda_n^2+t^2}{\lambda_n^2+1}\\ &=\sum_\rho\frac{1}{\frac{1}{2}+it-\rho}=\sum_{n=1}^{\infty}\frac{1}{i(t-\gamma_n)}=-i\sum_{n=1}^{\infty}\frac{1}{t-\gamma_n}\\ &=-i\,\mathrm{Tr}\left[(tI-\mathcal{O}_\alpha)^{-1}\right]=-i\sum_{n=1}^{\infty}\frac{1}{t-\lambda_n}\\ &=-i\frac{d}{dt}\log\det(tI-\mathcal{O}_\alpha),\qquad\rho=\frac{1}{2}+i\gamma_n,\end{aligned}\tag{147}$$

The KK spectrum of the PS model and the effective potential expansion (97) are connected, with all zeros γ_n appearing via (114) as poles of the resolvent

$$\begin{aligned}(\mathcal{O}_\alpha-i\gamma_n I)^{-1}\quad\text{has a simple pole at}\quad\gamma_n\quad&\Longleftrightarrow\quad\det(\mathcal{O}_\alpha-i\gamma_n I)=0\\ \Longleftrightarrow\quad\gamma_n\in\mathrm{spec}(\mathcal{O}_\alpha)\quad&\Longleftrightarrow\quad\zeta\Big(\frac{1}{2}+i\gamma_n\Big)=0,\qquad n\in\mathbb{N},\end{aligned}\tag{148}$$

The scattering platform in Sec. 3.7 relates zeros to reflection coefficient in (15). Near a zero, the impedance diverges as

$$\begin{aligned}Z(t)\sim\frac{1}{t-\gamma_n}=\frac{1}{\zeta'\big(\frac{1}{2}+i\gamma_n\big)\,(t-\gamma_n)}+\mathcal{O}(1)\sim\frac{1}{\zeta'\big(\frac{1}{2}+i\gamma_n\big)}\cdot\frac{1}{t-\gamma_n},\\ |Z(t)|\sim\frac{1}{\big|\zeta'\big(\frac{1}{2}+i\gamma_n\big)\big|}\cdot\frac{1}{|t-\gamma_n|}\sim\frac{1}{|t-\gamma_n|},\qquad t\to\gamma_n.\end{aligned}\tag{149}$$

The diffraction grating gives the same Fourier peaks at $\log p$ and $\log p^k$ as in (65), while the PF of the boundary CFT (104) has poles at the zeros

$$\begin{aligned}Z_{\text{CFT}}(\tau)\sim e^{\frac{\pi ic}{12\tau}}+\cdots\sim\exp\left[\frac{\pi ic}{12}\left(\frac{i}{2}+\frac{1}{\gamma_n}+\cdots\right)^{-1}\right]+\cdots\sim e^{-\frac{\pi c}{6}\gamma_n}+\cdots,\\ \tau_n=\frac{i}{2}+\frac{1}{\gamma_n}+\cdots,\end{aligned}\tag{150}$$

This provides a unified picture where Riemann zeros emerge as a physical spectrum connecting number theory, quantum chaos, gauge theory, string theory, and experimental optics.

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